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Scale-Free Dynamics: The Universal Somatic Field as a Complex Systems Framework

[T]-Theory Volume: Complex Systems and Emergence

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Introduction: The Same Equation at Every Scale

Complex systems science has a recurring embarrassment. The phenomena it studies — power-law distributions, critical transitions, scale-free networks, emergent coordination — appear at every scale of nature: in the firing patterns of neurons, in the flocking of birds, in the spread of epidemics, in the collapse of financial markets, in the dynamics of ecosystems, in the formation of galaxies. The mathematics that describes these phenomena is strikingly similar across contexts: the same scaling exponents, the same universality classes, the same renormalisation group flows. Yet the field lacks a principled account of *why* this similarity holds. The answer given is usually: universality — near-critical systems fall into universality classes determined by symmetry and dimensionality, and many natural systems happen to operate near criticality. But this answer explains the similarity of the scaling exponents without explaining why so many natural systems are near-critical in the first place.

This book presents a framework that offers a principled answer. The reason the same equations appear at every scale is that there is a single master equation — the Universal Somatic Field equation — that generates those dynamics at every scale as instantiations of the same underlying field theory. Scale-invariance is not an accident of proximity to criticality. It is a mathematical consequence of the compactification geometry of the field theory.

The 20-Scale Architecture

The USF framework defines a `ScaleUniverse` type covering 20 discrete scales, from quantum (scale 0) to cosmological (scale 19). At each scale, the somatic field equation takes the same form, with coupling constants that scale according to a specific power law determined by the compactification geometry. The power law is not fitted to data; it is derived from the moduli space of the compactification — the same mathematics that determines the Standard Model coupling constants in string phenomenology.

The consequence is a hierarchy of nested dynamics: the dynamics at scale $k + 1$ are the effective dynamics obtained by integrating out scale k. This is the renormalisation group picture applied to emotional and collective dynamics. The effective coupling at scale 7 (whole-brain) is determined by the coupling at scale 3 (single neuron) through a series of renormalisation group steps. The inter-personal field at scale 12 is determined by the individual fields at scale 7 through analogous steps.

This is not a metaphor for scale-invariance. It is a mathematical computation that can, in principle, be done explicitly — taking the coupling constants at the neural scale (measured from EEG or fMRI data) and running the renormalisation group equations up to the inter-personal scale, predicting the inter-personal coupling constants from first principles.

The Geographic Somatic Field

One of the most striking applications of the scale-invariance result is in the geographic domain. The geographic somatic field paper in this volume shows that the spread of dialect features across a linguistic geography, the formation of cultural identity boundaries, the dynamics of bird murmuration, and the synchronisation of firefly flashing all follow the same propagator equation — the somatic field Green's function evaluated at the appropriate scale. These are not coincidentally similar phenomena; they are the same phenomenon at different scales.

The practical implication for complex systems science is that existing models of geographic spread (reaction-diffusion equations, agent-based models, network models) are all approximations to the same underlying field equation. The USF framework provides the principled derivation from which these models can be recovered as limiting cases, and from which corrections can be calculated when the limiting assumptions break down.

Power Laws as Criticality Signatures

The USF framework gives a derivation — rather than an observation — of power-law statistics in complex systems. Near the somatic field phase transition (the consciousness threshold T_c, or the analogous criticality in social and ecological systems), the field fluctuations become scale-free: the correlation length diverges, and the fluctuation spectrum becomes a power law. The power-law exponent is determined by the universality class of the phase transition, which in turn is determined by the symmetry group of the somatic field.

This connects the USF framework to the self-organised criticality literature (Bak, Tang, Wiesenfeld) and to the neuronal avalanche literature (Beggs, Plenz). The difference is that the USF framework provides the field-theoretic foundation from which these results follow, rather than postulating criticality as a starting assumption. Systems self-organise to criticality because the somatic field dynamics are attractive toward the critical point: the energy landscape has a ridge structure that channels the dynamics toward the phase transition.

Gestalt Field Dynamics

The framework provides a formal account of Gestalt principles — the tendency of perception to organise local elements into coherent wholes — as field-theoretic phenomena. The Gestalt field dynamics paper in this volume shows that proximity, similarity, closure, and continuation — the classical Gestalt principles — correspond to specific features of the somatic field energy landscape: basin shape, coupling strength between adjacent modes, and the geometry of the saddle surfaces. The perception of a gestalt figure is the settlement of the perceptual field into an attractor basin corresponding to the coherent interpretation.

This connects the complex systems framework to the phenomenology of perception, and suggests that the Gestalt principles are not empirical generalisations (pattern-matching rules that happen to describe human perception) but necessary features of any field dynamics that satisfies the somatic field equation.

Emergence Without Magic

The central philosophical payoff of the complex systems perspective on the USF framework is a demystification of emergence. Emergence in complex systems is often described as the appearance of properties at higher scales that are not present at lower scales and cannot be predicted from lower-scale descriptions — *strong* emergence, in the philosophical terminology. The USF framework rejects strong emergence: the properties at higher scales are determined by the field equation at lower scales through the renormalisation group, and in principle they can be predicted.

What emergence *is*, in this picture, is the appearance of new *effective* degrees of freedom at higher scales — collective modes that are not present as individual constituents but arise from the integration of many lower-scale degrees of freedom. These effective degrees of freedom behave according to effective equations that look formally similar to the lower-scale equations (because the renormalisation group preserves the form of the equation) but with renormalised coupling constants. This is emergence without magic: real novelty at each scale, but novelty that is determined — not merely consistent with — the lower-scale dynamics.

What This Book Offers the Complex Systems Researcher

The papers assembled here develop the USF framework from the complex systems perspective: scale-invariance first (the zoomable somatic field paper), geographic dynamics next (the geographic somatic field paper), swarm coordination (the swarm propagator), and Gestalt field dynamics. The intended reader is comfortable with nonlinear dynamics, statistical mechanics, and network science, and need not have physics or neuroscience background.

Chapter 2 (zoomable somatic field) develops the `ScaleUniverse` type and the 20-scale renormalisation group. Chapter 3 (geographic somatic field) shows the empirical applications in geography, linguistics, and animal behaviour. Chapter 4 (swarm propagator) develops the $O(N^2)$ coordination result. Chapter 5 (Gestalt field dynamics) connects to perceptual psychology. The final chapter draws the research agenda: what measurements would test the renormalisation group predictions, and what the USF framework implies for the design of complex adaptive systems.

The same equation, every time. The question is: what does it imply at your scale?

Introduction

> *They saw a guitar string. I heard the music.*
>
> — A.J. (after Feynman)

The search for a unified description of physical reality has proceeded, since Newton, by identifying common mathematical structures across phenomena that appear superficially different. Fourier analysis revealed that the vibration of a string, the propagation of heat, and the conduction of electricity all obey the same equation with different boundary conditions. Maxwell showed that electricity and magnetism are aspects of a single field. Einstein showed that gravity and acceleration are locally indistinguishable.

The present work identifies a further unification: the same Green's function equation that governs electromagnetic propagation at the atomic scale (10^{-10} m) also governs seismic propagation at the geological scale (10^{5} m), cortical electromagnetic propagation at the neural scale (10^{-1} m), and gravitational wave propagation at the cosmological scale (10^{26} m). The wavenumber k changes at each scale; the equation does not.

This is not the observation that "waves are everywhere" — a qualitative truism — but a precise structural claim: the Green's function of any substrate with a characteristic oscillation frequency k satisfies the Helmholtz equation $(\nabla^2 + k^2)G = \delta$, and the solutions of this equation have the same form regardless of the physical medium. The propagator G is the medium's impulse response: its answer to the question *what happens at x given a unit perturbation at x'?*

This identification has a consequence for string theory. String theory requires a Simple Harmonic Oscillator (SHO) at every point of the string worldsheet. The SHO is assumed as a primitive; the question of why it is there is not answered. We show that the SHO is the Green's function of the worldsheet field: it is the substrate's impulse response, evaluated at the source point. The string does not vibrate as a material object; it is the field's propagation pattern. This is not a reinterpretation — it is a derivation from the structure of field equations.

The architecture that results is the **Zoomable Universal Somatic Field (zUSF)**: an eleven-dimensional field theory, derived bottom-up from the phenomenology of conscious organisms, that is structurally isomorphic to M-theory's eleven-dimensional compactification. The isomorphism is not metaphorical; it is a type-level proof verified by the Lean 4 kernel (`MTheoryIsomorphism.somaField_iso_mtheory`).

The derivation is inductive rather than deductive. Where Veneziano (1968) wrote down a scattering amplitude and Nambu, Nielsen, and Susskind separately identified the string as its underlying object, the present work identifies the Green's function as the oscillator. Where M-theory arrived at eleven dimensions from mathematical consistency requirements, the present work arrives at eleven by counting the functional degrees of freedom of a living organism. That the two derivations agree is the isomorphism at the heart of this paper.

Mathematical Foundation

2.1 The Master Equation

The foundational equation is the Helmholtz Green's function equation:

$$(\nabla^2 + k^2)\, G(x, x') = \delta(x - x') \tag{1}$$

where $x, x' \in \mathbb{R}^3$, $k > 0$ is the wavenumber, and δ is the Dirac delta distribution. Equation (1) admits the free-space solution:

$$G(x, x') = \frac{e^{ik|x-x'|}}{4\pi|x - x'|} \tag{2}$$

This is the retarded propagator: the field amplitude at x due to a unit point source at x'. Three properties are immediate:

1. **Positivity of amplitude**: $|G| > 0$ for all $x \neq x'$.
2. **Decay**: $|G| \sim 1/r$ as $r = |x - x'| \to \infty$ (radiation condition).
3. **SHO structure**: For fixed x, the function $x' \mapsto G(x, x')$ satisfies $(\partial^2/\partial {x'}^2 + k^2)G = \delta$ — the harmonic oscillator equation in the source variable.

Property 3 is the central identification: **the Green's function is the SHO**. The SHO of string theory, required at every worldsheet point, is the substrate's impulse response. This observation, formalised in the companion file `UniversalSomaticField.lean` (axiom `greens_fn_is_SHO`), is the structural core of the zUSF.

2.2 Scale Invariance

As k varies from $k_P = \ell_P^{-1} \approx 10^{35}\ \mathrm{m}^{-1}$ (Planck scale) to $k_H = \ell_H^{-1} \approx 10^{-26}\ \mathrm{m}^{-1}$ (Hubble scale), the form of equation (1) is invariant. The boundary conditions and the physical interpretation of G change; the equation does not.

Definition (Scale-Invariant Field). A field G is *scale-invariant* if for every scale $\sigma \in \{0, \dots, 20\}$ there exists a wavenumber $k(\sigma) > 0$ and a physical substrate $\mathcal{S}(\sigma)$ such that G satisfies equation (1) with those parameters.

Theorem (Lean 4 verified, `UniversalSomaticField.universal_field_theory`): G is scale-invariant across all 20 levels.

The 20-step scale dial: each level coloured from violet (Planck) to yellow (cosmic). The master equation $(\nabla^2 + k^2)G = \delta$ is invariant across all levels; only k changes. (*Generated:* `FA_universal_dial.png`)

Figure 1: The 20-step scale dial: each level coloured from violet (Planck) to yellow (cosmic). The master equation $(\nabla^2 + k^2)G = \delta$ is invariant across all levels; only k changes. (*Generated:* `FA_universal_dial.png`)

Correspondence Principle: softmax$(+1)$ converges to 1 as $\beta \to \infty$ (log scale). At $\beta = 50$, the output is numerically indistinguishable from the classical sign function. (*Generated:* `FSx_softmax_correspondence.png`)

Figure 2: Correspondence Principle: softmax$(+1)$ converges to 1 as $\beta \to \infty$ (log scale). At $\beta = 50$, the output is numerically indistinguishable from the classical sign function. (*Generated:* `FSx_softmax_correspondence.png`)

Proof. By `scale_invariance_inhabited`: for every σ, the type `FieldEquation σ` is inhabited. □

2.3 Log-Sum-Exp and the Correspondence Limit

For the biological substrate at Scale 6, the propagator takes a modified form. The FM-HN architecture (§7) uses the log-sum-exp energy:

$$E_{20}(\xi) = -\frac{1}{\beta} \log \sum_{\mu} e^{\beta \xi^{\mu T} \xi} + \frac{1}{2}\|\xi\|^2 \tag{3}$$

whose update rule $\xi \leftarrow X^T \cdot \mathrm{softmax}(\beta \cdot X\xi)$ converges to the classical sign update as $\beta \to \infty$:

$$\lim_{\beta \to \infty} \mathrm{softmax}(\beta \cdot z)_i = \mathbf{1}[i = \arg\max z] \tag{4}$$

This limit — the Correspondence Principle — is verified in `LimbicHopfield.correspondence_principle` and illustrated in figure 2.

The Eleven-Dimensional Architecture

3.1 Decomposition

Let $\mathcal{M}_{11}$ denote the configuration space of a living organism in interaction with its environment. We decompose $\mathcal{M}_{11}$ as:

$$\mathcal{M}_{11} = \underbrace{M_4}_{\text{Spacetime}} \times \underbrace{P_3}_{\text{Propagator}} \times \underbrace{L_1}_{\text{Limbic}} \times \underbrace{C_3}_{\text{Cortex}} \tag{5}$$

The four subspaces are:

Symbol	Dim	Physical substrate	Mathematical role
M_4	4	Body in 3+1D spacetime	Lorentzian manifold, causal structure
P_3	3	Endogenous EMF (CEMI field)	Green's function domain
L_1	1	Limbic axis (homeostatic regulation)	Orbifold segment $[-1, 1]$
C_3	3	Cortical information routing	Green's function co-domain

The compact 7-dimensional internal space is:

$$X_7 = P_3 \times L_1 \times C_3, \quad \dim(X_7) = 3 + 1 + 3 = 7 \tag{6}$$

Theorem (Lean 4 verified, `MTheoryIsomorphism.dim_is_11`): $4 + 3 + 1 + 3 = 11$. Proof by `decide`. □

3.2 Isomorphism with M-Theory

M-theory (Witten 1995) compactifies eleven-dimensional supergravity as $M_{11} = M_4 \times X_7$ where X_7 is a compact manifold with G_2 holonomy. The soma-field decomposition (5) has identical dimensional structure.

Theorem (Lean 4 verified, `MTheoryIsomorphism.somaField_iso_mtheory`): There exists a type isomorphism:

$$\text{SomaField}_{11} \cong \text{Spacetime} \times \text{CompactSpace}_7 \tag{7}$$

Proof. By `toMTheory` and `fromMTheory`; roundtrip by `simp`. □

The derivation is independent: the M-theory structure was not assumed; it was arrived at by counting functional degrees of freedom of a biological system. The isomorphism (7) is therefore a theorem, not a construction.

3.3 The Limbic Axis as a Horava-Witten Orbifold

In Horava-Witten M-theory (1996), the compact direction is an orbifold $S^1/\mathbb{Z}_2$ — a line segment with two ten-dimensional boundary spacetimes at each endpoint. The Limbic Axis $L_1 \cong [-1, 1]$ has the same structure:

- Endpoint $x = -1$: somatic boundary (body-world)
- Endpoint $x = +1$: cortical boundary (mind-world)

The quartic double-well potential $V(x) = W(x^2 - 1)^2$ for barrier heights $W \in \{8, 10, 12\}$ corresponding to the QUANT-EXP-1 sweep. (*Generated:* `FS0_double_well.png`)

Figure 3: The quartic double-well potential $V(x) = W(x^2 - 1)^2$ for barrier heights $W \in \{8, 10, 12\}$ corresponding to the QUANT-EXP-1 sweep. (*Generated:* `FS0_double_well.png`)

- Interior $(-1, 1)$: transition zone, subject to quantum tunnelling

Theorem (Lean 4 verified, `MTheoryIsomorphism.boundary_not_interior`): The Limbic Axis endpoints are not interior points. □

The double-well potential on L_1:

$$V(x) = W(x^2 - 1)^2 \tag{8}$$

models the energy barrier between somatic and cortical attractors. At $x = -1$ (trauma attractor), classical gradient descent is trapped: `LimbicTunnel.gradient_traps_near_neg1` (proved by `nlinarith`).

The Zoom Operator

4.1 Definition

Definition (Zoom Operator). The Zoom Operator Λ is a dependent type constructor:

$$\Lambda : \text{ScaleLevel} \to \text{FieldEquation} \tag{9}$$

where ScaleLevel $= \text{Fin}(21)$ and FieldEquation(σ) packages the wavenumber $k(\sigma)$, boundary conditions, and substrate type.

In Lean 4:

```
structure FieldEquation (n : ScaleLevel) where
  k : □
  hk : 0 < k
  G : □ → □ → □
```

Remark on the choice of 20 levels. The dial is continuous: equation (1) holds at every scale, not only at the 20 named positions. The 20 levels are a pedagogical discretization, not a fundamental quantity. The choice is motivated by two considerations. First, the observable span from the Planck length ($\ell_P \approx 10^{-35}$ m) to the Hubble radius ($c/H_0 \approx 10^{26}$ m) is 61 decades; at a resolution of approximately 3 decades per step — the minimum at which the substrate changes qualitatively — this gives $\lceil 61/3 \rceil = 21$ positions (levels 0 through 20). Second, the 20 steps coincide with five major qualitative phase transitions at which the governing physics changes character, each spanning approximately four steps:

Transition	Levels	Nature of change
Quantum → Classical	0–4	Spacetime geometry emerges; probability amplitude collapses to matter
Chemistry → Biology	4–7	Self-replication and homeostatic regulation appear
Individual → Collective	7–10	Agency distributes across coupled agents
Geological → Stellar	10–14	Self-gravity dominates over chemical binding
Stellar → Cosmic	14–20	Dark energy and expansion compete with gravity

The number 20 is therefore not arbitrary, but it is also not uniquely determined. A discretization into 15 or 25 steps would be equally defensible. The scientific claim is about the invariance of equation (1), not about the count of steps. The steps are tick marks on a continuous dial.

4.2 Physical and Mind Scaling

Physical scaling proceeds through the characteristic length $\ell(\sigma) \sim k(\sigma)^{-1}$, ranging from 10^{-35} m $(\sigma = 0)$ to 10^{26} m $(\sigma = 20)$.

Mind scaling proceeds through the tensor rank $N(\sigma)$:

$$N(\sigma) \approx \begin{cases} 10^2 & \sigma = 2 \text{ (nuclear)} \\ 10^{14} & \sigma = 6 \text{ (brain)} \\ 10^{11} & \sigma = 15 \text{ (galactic)} \\ \infty & \sigma = 20 \text{ (universal)} \end{cases} \tag{10}$$

Theorem (architectural constraint): Physical scale $\ell(\sigma)$ and mind rank $N(\sigma)$ zoom together. They cannot zoom independently.

Proof. The Zoom Operator returns a dependent pair $(\ell(\sigma), N(\sigma))$ whose components are both functions of the same σ. Setting σ determines both simultaneously. □

Physical scale (left, log metres) and mind rank N (right, log units) both increase with σ from 0 to 20. The two bars are tethered — a change in one forces a change in the other. (*Generated:* `FA_dual_scaling.png`)

Figure 4: Physical scale (left, log metres) and mind rank N (right, log units) both increase with σ from 0 to 20. The two bars are tethered — a change in one forces a change in the other. (*Generated:* `FA_dual_scaling.png`)

The Twenty-Scale Catalogue

This section instantiates equation (1) at each of the twenty scale levels. For each level we state: the physical substrate, the Green's function interpretation, the mind matrix, and the equation parameters. The pattern is invariant: only the labels change.

Scale 0 — Quantum Foam (10^{-35} m)

Equation parameters: $k = k_P = \ell_P^{-1} \approx 10^{35}\ \mathrm{m}^{-1}$; boundary: periodic (no preferred direction); $N = \infty$ (all configurations in superposition).

Physical substrate: Discrete spacetime nodes; pre-geometric fluctuations. At the Planck scale, geometry itself becomes probabilistic. The metric fluctuates with amplitude $\delta g \sim 1$ (Planck units).

Propagator: G is the gravitational quantum amplitude — the probability amplitude for a graviton to propagate from x' to x. In the semiclassical limit: $G_P(x, x') = \langle x|\hat{G}|x'\rangle$ (Feynman path integral). The worldsheet SHO is G evaluated at the string scale (Scale 1).

Mind matrix: Quantum superposition state. The S-matrix encodes all possible scattering outcomes as complex amplitudes. $N_0 = \dim(\mathcal{H}_{\text{Planck}}) = \infty$.

Remark. Scale 0 is where equation (1) takes its most abstract form. Every subsequent scale is this equation, coarse-grained.

Scale 1 — String Scale (10^{-32} m)

Equation parameters: $k = \ell_s^{-1} \approx 10^{32}\ \mathrm{m}^{-1}$; boundary: periodic (closed string) or Dirichlet (open string on D-brane).

Physical substrate: String worldsheets; M-theory 2-branes. The string characteristic length is $\ell_s \approx 10^{-32}$ m.

Propagator: The worldsheet Green's function: $G_{\text{string}}(\sigma, \sigma') = -\frac{\alpha'}{2} \ln|\sigma - \sigma'|^2$ (free bosonic string, Regge slope α'). The vibrational modes satisfy the SHO equation $\ddot{X}^n + n^2 X^n = 0$.

Left: the Simple Harmonic Oscillator ($\ddot{x} + \omega^2 x = 0$). Right: the Green's function $G(\tau)$ of a harmonic system — both satisfy the SHO equation. They are the same object. (*Generated:* `FS1_sho_string.png`)

Figure 5: Left: the Simple Harmonic Oscillator ($\ddot{x} + \omega^2 x = 0$). Right: the Green's function $G(\tau)$ of a harmonic system — both satisfy the SHO equation. They are the same object. (*Generated:* `FS1_sho_string.png`)

Yukawa potential e^{-mr}/r (nuclear, solid) versus Coulomb potential $1/r$ (electromagnetic, dashed). Same master equation; different mass parameter k. (*Generated:* `FS2_yukawa_vs_coulomb.png`)

Figure 6: Yukawa potential e^{-mr}/r (nuclear, solid) versus Coulomb potential $1/r$ (electromagnetic, dashed). Same master equation; different mass parameter k. (*Generated:* `FS2_yukawa_vs_coulomb.png`)

Key result. The SHO of string theory IS G. A string vibrational mode at frequency n is the n-th Fourier mode of the worldsheet's impulse response. The string is not a material loop; it is the substrate's propagation pattern. This is `UniversalSomaticField.greens_fn_is_SHO` (axiom).

Mind matrix: The string landscape: $N \sim 10^{500}$ vacuum configurations. Each selects a different low-energy physics. Our universe occupies one vacuum.

Scale 2 — Nuclear (10^{-15} m)

Equation parameters: $k = m_\pi c/\hbar \approx 10^{15}\ \text{m}^{-1}$; boundary: confinement radius $r \lesssim 1$ fm.

Physical substrate: Quarks, gluons, atomic nuclei. Strong nuclear force confines quarks; the residual strong force between nucleons is mediated by pion exchange.

Propagator: Yukawa kernel: $G_{\text{nuc}}(r) = e^{-m_\pi r}/(4\pi r)$. The exponential factor $e^{-m_\pi r}$ encodes finite range. Setting $m_\pi = 0$ recovers the Coulomb propagator of Scale 3 — the transition from a massive to a massless carrier.

Mind matrix: Nuclear S-matrix ($N \approx 10^5$ nuclear energy levels across all stable nuclei). Binding energy curve = eigenvalue spectrum of G_{nuc}.

Scale 3 — Atomic (10^{-10} m)

Equation parameters: $k = \sqrt{2m_e E}/\hbar$; boundary: molecular orbital extent; $k = 0$ for the static Coulomb case.

Physical substrate: Atoms; electron orbitals; covalent bonds. The characteristic length is the Bohr radius $a_0 = 0.529$ Å.

Propagator: Coulomb kernel: $G_{\text{EM}}(r) = 1/(4\pi r)$. This is the $k \to 0$ limit of equation (2): the photon is massless, giving infinite range. The Schrödinger equation with Coulomb potential generates atomic orbitals as eigenfunctions of G_{EM}.

Key property. The first infinite-range propagator in the catalogue. Electromagnetism reaches across the observable universe.

Mind matrix: Atomic orbital basis ($N \approx 10^2$ per atom). Ionisation energy = barrier height of the atomic attractor.

Same as always. The $1/r$ dependence of G_{EM} at atomic scale (10^{-10} m) is identical in form to the gravitational propagator at cosmological scale (10^{26} m). Equation (1) with $k = 0$.

Scale 4 — Molecular (10^{-9} m)

Equation parameters: $k = \sqrt{2m_e E_{\text{bond}}}/\hbar$; boundary: nuclear positions and molecular geometry.

Physical substrate: Chemical bonds; crystal lattices; biological macromolecules (proteins, DNA). Molecular geometry is the ground state of the electron density under nuclear boundary conditions.

Propagator: Schrödinger Green's function: electron density amplitude $G_e(x, x') = e^{ik|x-x'|}/(4\pi|x-x'|)$. Molecular orbitals are eigenmodes of G_e — resonances of the propagator under nuclear constraints.

Mind matrix: Molecular conformational space ($N \approx 10^3$ per protein). Protein folding = energy minimisation on the molecular attractor landscape. A conformational transition (e.g., retinal 11-cis → all-trans, triggering vision) is an attractor transition in G_e.

Same as always. The molecular conformation double-well $V(x) = W_{\text{mol}}(x^2 - 1)^2$ with $W_{\text{mol}} \approx 4$ eV is identical in structure to the limbic double-well (Scale 6, $W \approx 10$ natural units) — separated by twenty-five orders of magnitude in characteristic length.

Scale 5 — Cellular / Neural (10^{-6} m)

Equation parameters: $k = \lambda_{\text{axon}}^{-1} \approx 2000\ \text{m}^{-1}$ (axon space constant $\lambda \approx 0.5$ mm); $N \approx 10^4$ per neuron.

Physical substrate: Neurons; synapses; axon fibres; fascial networks. The cable equation $(\partial^2/\partial x^2 - \lambda^{-2})V = I_{\text{inj}}$ is the hyperbolic form of equation (1) with $k = i\lambda^{-1}$.

Propagator: Synaptic transfer function. Neural impulse response encodes how a spike at the pre-synaptic terminal propagates to the post-synaptic membrane. Ephaptic coupling (Anastassiou et al. 2011) extends this to the electric field generated by synchronised neural firing.

Mind matrix: Synaptic weight matrix W_{ij} (Hopfield 1982). Stored memories are local minima of $E_{82}(s) = -\frac{1}{2}s^T W s$. Storage capacity: approximately $0.14 \cdot D$ patterns for D-dimensional state space.

Key result. QUANT-EXP-1 (Johnson, 2026b): quantum annealing achieves escape from the trauma attractor (barrier $W \in \{8, 10, 12\}$) with 3/3 success rate; classical Langevin dynamics achieve 0/48. WKB tunnelling amplitude:

$$\Theta(W) = \exp\left(-\frac{8\sqrt{2W}}{3}\right) > 0 \quad \forall W > 0 \tag{11}$$

(proved: `LimbicTunnel.wkbAmplitude_pos`).

Scale 6 — Brain / CEMI Field (10^{-1} m)

Equation parameters: $k = \omega/c_{\text{neural}} \approx 2\pi \times 40\ \text{Hz}/6\ \text{m s}^{-1} \approx 40\ \text{m}^{-1}$ (gamma band); $N \approx 10^{14}$ (synaptic connections).

Physical substrate: Cerebral cortex; subcortical nuclei; 86 billion neurons; approximately 10^{14} synapses organised into cortical layers.

Propagator: Conscious Electromagnetic Information (CEMI) field (McFadden, 2002a, 2002b): the macroscopic electromagnetic field generated by synchronised neural firing across the cortex. Measurable by magnetoencephalography (MEG) and magnetocardiography (MCG) at the body surface. Field feeds back onto neuronal firing thresholds (ephaptic gain), producing a self-modulating propagator.

Mind matrix: Subjective awareness and associative memory. The Hopfield energy landscape maps traumatic configurations to deep attractor basins; the FM-HN architecture (§7) provides the runtime coupling to the limbic field.

Consciousness threshold: Awareness emerges when the CEMI field amplitude $\phi \geq T_c = \sqrt{2}$ (normalised units). Proved: `UniversalSomaticField.consciousness_dichotomy`.

Scale 7 — Organism (10^0 m)

Equation parameters: $k = \omega / c_{\text{tissue}}$ (c_{tissue}: speed of elastic waves in fascia and soft tissue); $N \approx 10^{14}$–10^{15}.

Physical substrate: The body as a biotensegrity structure (Ingber 1998): a pre-stressed, globally coupled elastic network of skeleton, fascia, muscle, and viscera. Not a stack of parts but a continuous wave-bearing medium.

Propagator: Full somatic CEMI field (the complete 11D configuration space of equation (5)). The cardiac electromagnetic field — the loudest electromagnetic event the body produces — is detectable by MCG at distances up to 2 m from the body surface.

Mind matrix: The full 11D organism; all four subspaces active. Subjective experience, emotional regulation, trauma, creativity. The FM-HN architecture (§7) governs runtime dynamics.

Scale 8 — Animal Swarms (10^0–10^1 m)

Equation parameters: $k \sim r_{\text{align}}^{-1}$ (alignment radius); N = swarm size.

Physical substrate: Discrete agents (birds, fish, insects, drones) in 3-dimensional space, each responding to local neighbours.

Propagator: Active-matter velocity field (Toner and Tu 1995): $\partial_t \mathbf{v} + \lambda(\mathbf{v} \cdot \nabla)\mathbf{v} = -\nabla P + D_T \nabla^2 \mathbf{v}$. Global formation emerges from local interactions propagated through the swarm by the same Green's function structure as all preceding scales.

Key result (swarm coordination, P16 (Johnson, 2026c)): Treating the swarm as a macroscopic brane projection reduces coordination cost from $O(N \cdot K)$ to $O(N^2)$ with $K = 1$. The Green's function replaces K rounds of message-passing with a single matrix-vector product. Jam resistance follows as a corollary ($K = 1$ means no communication round to disrupt). Proved: `SwarmPropagator.propagator_beats_classical`, `SwarmPropagator.jam_resistant`.

Scale 9 — Society / City (10^3 m)

Equation parameters: $k = r_{\text{interaction}}^{-1} \approx 10^{-3}\ \text{m}^{-1}$; $N \approx 10^6$–10^7 (city population).

Physical substrate: Urban infrastructure; transport networks; population distribution. The city as a physical coupling network.

Propagator: Social interaction kernel G_{ij} — how frequently agent i encounters agent j in the physical medium. Cultural propagation (dialect spread, technology adoption) is a structural contagion wave governed by the social Green's function: $P(s_i \to 1) = \sigma(\sum_j G_{ij}s_j - \theta)$ (social Hopfield network).

Mind matrix: Cultural attractors ($N \approx 10^3$ stable cultural modes). Language variants, norms, and fashions are attractor states of the social field. Estuary English propagation along the Thames Valley is a worked example of geographic boundary conditions selecting which modes propagate and which decay (§12).

Scale 10 — Geological (10^5 m)

Equation parameters: $k = \omega/v_P \approx \omega/(6000\,\text{m/s})$ (P-wave velocity); boundary: crustal moho below, free surface above.

Physical substrate: Tectonic plates; the Alpine fold-and-thrust belt; the Klöntalersee basin (Glarus, Switzerland) as a natural acoustic resonator. The Glarus Hauptüberschiebung places Verrucano sandstone (250 Ma) over Eocene flysch (35 Ma), recording 35 km of northward transport — a wave with a ten-million-year period.

Propagator: Seismic Green's function, measurable by global seismometer networks. The Earth's normal modes (free oscillations) are eigenstates of the elastic Green's function under spherical boundary conditions.

Mind matrix: Crustal stress distribution tensor ($N \approx 10^3$ stress modes). Geological memory: the fold geometry of a mountain range encodes every collision the lithosphere has experienced. The rock face is a four-dimensional document read as a three-dimensional spatial slice.

Scale 11 — Planetary (10^6 m)

Equation parameters: Navier-Stokes + heat equation in a rotating frame; effective k set by thermodynamic convection wavelengths.

Physical substrate: Planetary mantle and core. The mantle convects on timescales of millions of years under gravitational and thermal forcing. The geodynamo (liquid outer core) generates the planetary magnetic field.

Propagator: Thermodynamic convection kernel; seismic tomography provides the empirical Green's function for the Earth's interior.

Mind matrix: Global energetic equilibrium; carbon cycle; ice-age attractor sequence. The climate system is a dynamical system with at least two stable attractors (glacial and interglacial) separated by a bifurcation controlled by orbital forcing (Milankovitch cycles).

Scale 12 — Orbital (10^9 m)

Equation parameters: Newtonian gravity; $k \to 0$ (long-range, massless graviton).

Physical substrate: Planetary and lunar orbits; the heliosphere; Lagrange points. Solar wind creates an effective medium with frequency-dependent propagation properties.

Propagator: Gravitational Coulomb kernel $G_{\text{grav}}(r) = -Gm/r$ (same $1/r$ form as the electromagnetic Coulomb kernel of Scale 3, with a different coupling constant and sign). Gravitational lensing provides a direct measurement of G_{grav}.

Mind matrix: Orbital resonance structure. The solar system's Kirkwood gaps and mean-motion resonances are the stable eigenstates of the gravitational N-body problem.

Scale 13 — Stellar (10^{11} m)

Equation parameters: $k = \omega/c_s$ (sound speed in stellar plasma $c_s \approx 100$ km/s); $N \approx 10^6$ oscillation modes.

Physical substrate: Stars: thermonuclear plasma in hydrostatic equilibrium, bounded by radiation pressure and gravity.

Propagator: Helioseismic Green's function, directly measured by the Solar Dynamics Observatory. The Sun supports approximately 10^7 simultaneous acoustic modes (p-modes, g-modes, f-modes) spanning 5 minutes to hours.

Mind matrix: Stellar oscillation spectrum. The eigenfrequency distribution encodes the interior structure — density profile, rotation, chemical stratification. Asteroseismology reads the mind matrix of distant stars.

Scale 14 — Black Holes and Compact Objects (10^3–10^{10} m)

Equation parameters: $k = 2\pi f_{\text{ISCO}}/c$ (innermost stable circular orbit frequency); boundary: event horizon.

Physical substrate: Neutron stars; black holes; binary inspiral systems. The extreme curvature regime of general relativity.

Propagator: Quasi-normal modes — the gravitational wave propagator for a perturbed black hole. Each quasi-normal mode is a damped oscillation: $G_{\text{BH}}(t) \propto e^{-t/\tau_{\text{ring}}} \cos(\omega_{\text{QNM}} t)$. This is the impulse response of curved spacetime.

Mind matrix: Black hole thermodynamic state (Bekenstein entropy $S = A/4G\hbar$, where A is the horizon area). The Bekenstein-Hawking entropy encodes $\sim 10^{77}$ bits for a solar-mass black hole.

Scale 15–16 — Galactic (10^{20}–10^{22} m)

Equation parameters: Poisson-Vlasov system; $k \sim \pi/R_{\text{arm}}$ (spiral arm half-wavelength).

Physical substrate: Stellar populations ($\sim 10^{11}$ stars per galaxy); dark matter halo; interstellar medium.

Propagator: Density-wave kernel (Lin and Shu 1964). Spiral arms are not physical structures of permanently bound stars; they are density waves — compressions propagating through the stellar fluid. The Green's function of the galactic disk determines which pattern speeds are stable.

Mind matrix: Galactic kinematics; rotation curve; spiral arm pattern. $N \approx 10^{11}$ (number of stars in the Milky Way). The rotation curve encodes the mass distribution including dark matter.

Scale 17–18 — Large-Scale Structure (10^{23}–10^{24} m)

Equation parameters: Linearised cosmological perturbation theory; $k \sim k_{\text{BAO}} = 0.1\,\text{Mpc}^{-1}$ (baryon acoustic oscillation scale).

Physical substrate: Galaxy clusters; cosmic filaments; voids. The large-scale structure traces the initial density perturbations from inflation, propagated through the baryon-photon fluid before recombination.

Propagator: Baryon Acoustic Oscillation (BAO) kernel: the Green's function of acoustic waves in the primordial plasma, imprinted on the matter distribution at recombination. BAO provides a standard ruler for cosmological distance measurement.

Mind matrix: Large-scale structure topology; cosmic web connectivity. $N \approx 10^{14}$ (number of galaxies in the observable universe).

Scale 19–20 — Observable Universe (10^{26} m)

Equation parameters: Linearised Einstein equation $\Box h_{\mu\nu} = -16\pi G T_{\mu\nu}$; $k = \omega/c$; $N \to \infty$.

Physical substrate: The full observable universe from the surface of last scattering ($z = 1100$) to the Hubble sphere ($c/H_0 \approx 4.4$ Gpc). Dark energy dominates the current energy budget.

Propagator: Gravitational wave propagator — the retarded Green's function of the linearised Einstein equation: $G_{\mathrm{GW}}(x, x') = \theta(t - t')\delta((x - x')^2)/(2\pi)$. This is spacetime's impulse response. Gravity is the Green's function of the metric field. LIGO/Virgo/KAGRA measure G_{GW} directly.

Mind matrix: The global cosmological state. If the universal CEMI field amplitude satisfies $\phi_{\mathrm{cosmic}} \geq T_c$, the universe satisfies the structural requirements for consciousness (`UniversalSomaticField.universe_is_11D_organism`, axiom). Whether this condition is met dynamically is an empirical question.

Same as always. The gravitational wave propagator at Scale 20 is formally identical to the Coulomb propagator at Scale 3 (both are $1/r$ forms of equation (1) with $k = 0$) and to the synaptic transfer function at Scale 5. One equation. Twenty scales.

Consciousness as Phase Transition

6.1 Definition

Definition (Pre-conscious state). A system at Scale 6–7 is *pre-conscious* when its limbic field amplitude $\phi < T_c$. Field propagation occurs; no first-person awareness is present.

Definition (Conscious state). A system is *conscious* when $\phi \geq T_c$. The limbic field couples the somatic and cortical subspaces; first-person awareness emerges as a property of this coupling.

Theorem (Lean 4 verified, `UniversalSomaticField.consciousness_dichotomy`): For any $\phi \in \mathbb{R}$, either $\phi < T_c$ (pre-conscious) or $\phi \geq T_c$ (conscious). The transition is sharp. □

Theorem (Lean 4 verified, `UniversalSomaticField.consciousness_monotone`): Raising ϕ cannot destroy consciousness. □

6.2 The Hard Problem

The "hard problem of consciousness" (Chalmers 1995) asks why physical processes give rise to subjective experience. On the present account, the question is reframed: *why does the limbic field amplitude exceed T_c?* This is an empirical question about field dynamics, not a philosophical puzzle about the relation between matter and mind.

WKB tunnelling amplitude $\Theta(W)$ vs. barrier height W. QUANT-EXP-1 values $(W = 8, 10, 12)$ marked. Classical rate = 0; quantum rate = $\Theta > 0$ always. (*Generated:* `FS6_wkb_amplitude.png`)

Figure 7: WKB tunnelling amplitude $\Theta(W)$ vs. barrier height W. QUANT-EXP-1 values $(W = 8, 10, 12)$ marked. Classical rate = 0; quantum rate = $\Theta > 0$ always. (*Generated:* `FS6_wkb_amplitude.png`)

A conscious percept is a **pole in the propagator**: the field's first-person experience of its own impulse response, occurring when the excitation frequency matches a natural resonance of the manifold. The "felt quality" (quale) is the resonance; the "content" is the mode structure.

6.3 The Trauma Attractor

Trauma is a topological obstruction: a configuration of the limbic field L_1 with a high-barrier double well. Classical gradient descent cannot escape:

$$\frac{dx}{dt} = -V'(x) = -4Wx(x^2 - 1) < 0 \quad \text{for } x \in (-1, 0)$$

This traps the system near $x = -1$ indefinitely. Quantum tunnelling provides the only escape route. The WKB amplitude:

$$\Theta(W) = \exp\left(-\frac{8\sqrt{2W}}{3}\right) \tag{12}$$

is strictly positive for all finite W (proved: `LimbicTunnel.wkbAmplitude_pos`) but exponentially small for large W. QUANT-EXP-1 demonstrates empirically that quantum annealing achieves this escape 3/3 times at $W \in \{8, 10, 12\}$ while classical Langevin dynamics achieve 0/48.

The Field-Modulated Hopfield Network

7.1 Architecture

The FM-HN ((Johnson, 2026d)) unifies the classical 1982 Hopfield network (Hopfield, 1982) and the modern 2020 network (Ramsauer et al., 2020) as limiting cases of a single architecture parameterised by the limbic field amplitude Φ.

Two runtime coupling equations:

$$T(t) = T_0 + \sigma \cdot \Phi_{\text{limbic}}(t) \tag{13}$$

$$W(t) = W_0 + \gamma \cdot \Phi_{\text{limbic}}(t) \cdot J \tag{14}$$

where $T_0 > 0$ is the baseline temperature, $\sigma > 0$ the limbic coupling strength, $J \in \mathbb{R}^{D\times D}$ the coupling matrix, and $\gamma > 0$ the ephaptic gain coefficient.

7.2 Correspondence Principle

Theorem (Lean 4 verified, `LimbicHopfield.correspondence_principle`): Under zero somatic stress $\Phi = 0$:

$$T(t) = T_0, \quad W(t) = W_0 \tag{15}$$

Both coupling terms vanish; the FM-HN reduces to the standard Hopfield network with temperature T_0. As $T_0 \to 0$ ($\beta \to \infty$):

$$\text{FM-HN update} \to \text{sign}(W_0 \cdot s) = \text{HN-1982 update}$$

Proof: `calm_temp_is_baseline` and `calm_weight_is_baseline` by `simp`. □

This is the Einstein-Newton relationship for neural architectures: the 1982 network is the low-temperature, calm-somatic limit of the FM-HN, just as Newtonian mechanics is the low-velocity limit of special relativity.

7.3 Neurodivergent Operator Modifications

The FM-HN parameter space (β, W) contains distinct regimes corresponding to neurodivergent profiles (all proved by `linarith` in `LimbicHopfield`):

Profile	Baseline T	Barrier W	Dynamical regime
ADHD	$1.8 \cdot T_0$ (hot)	Low	Rapid exploration, low settling
ASC	$0.4 \cdot T_0$ (cold)	Normal	Deep attractors, rare transitions
C-PTSD	T_0	High ($W = 12$)	Classical trapping, quantum escape needed

Theorem (Lean 4 verified, `LimbicHopfield.adhd_hotter_than_autism`): $T_{\text{ASC}} < T_0 < T_{\text{ADHD}}$. □

The Arnold tongue: stable frequency-locked region (shaded) in the parameter space of coupling strength vs. frequency detuning. Rapport = operating inside the tongue. (*Generated:* `FS8_arnold_tongue.png`)

Figure 8: The Arnold tongue: stable frequency-locked region (shaded) in the parameter space of coupling strength vs. frequency detuning. Rapport = operating inside the tongue. (*Generated:* `FS8_arnold_tongue.png`)

The Relational Field

When two organisms interact, the 11D decomposition extends to a coupled system. The single-organism propagator $G \in \mathbb{R}^{N\times N}$ becomes a block matrix:

$$\mathbf{G}_{AB}(\omega) = \begin{pmatrix} G_{AA}(\omega) & G_{AB}(\omega) \\ G_{BA}(\omega) & G_{BB}(\omega) \end{pmatrix} \tag{16}$$

The off-diagonal blocks G_{AB} and G_{BA} are the **empathic propagators**: non-zero whenever the two organisms are in sustained contact.

Huygens frequency locking. Two coupled oscillators with natural frequencies ω_A and ω_B and coupling $\kappa = |G_{AB}|$ lock to a common frequency when:

$$|\omega_A - \omega_B| < \kappa \tag{17}$$

(Arnold tongue condition). Rapport is the phenomenological signature of frequency locking. The Arnold tongue width grows with friendship depth (persistent G_{AB}), explaining why close relationships synchronise across longer frequency separations.

Therapist-client entrainment. The therapist's regulated field (low W_T, stable attractor) modifies the client's effective barrier via:

$$W_{\text{eff}} = W \cdot (1 - \alpha |G_{TC}|^2) \tag{18}$$

As therapeutic alliance deepens ($|G_{TC}|^2$ grows), W_{eff} decreases and the WKB tunnelling amplitude $\Theta(W_{\text{eff}})$ increases. This provides a quantitative prediction: the depth of therapeutic alliance should predict the rate of symptomatic improvement in trauma-spectrum conditions.

Encapsulation of Related Frameworks

The zUSF encapsulates three existing frameworks as special cases or scale-restricted projections.

9.1 McFadden's CEMI Theory (McFadden, 2002a, 2002b)

McFadden proposes that consciousness correlates with the brain's endogenous electromagnetic field. In the zUSF: the CEMI field is the Scale-6 ($\sigma = 6$) restriction of the universal propagator G. The zUSF extends CEMI in two directions: downward to quantum neural noise (Scale 5) and upward to multi-organism coupling (§8) and cosmological propagation (Scale 20).

9.2 Schreiber's Modal Homotopy Type Theory

Schreiber (2013) formalises physics in dependent type theory, arriving at an 11-dimensional structure from the mathematics of M-theory. The zUSF arrives at the same 11-dimensional structure from the bottom up (clinical observation). The structural isomorphism (theorem 3.2) confirms that the two approaches describe the same object. The zUSF provides the biological execution engine that Schreiber's purely mathematical framework lacks.

9.3 Hoffman's Conscious Agents Model (Hoffman, 2019)

Hoffman proposes that spacetime is a "user interface" constructed by conscious agents; it is not fundamental. The zUSF disagrees on one point: spacetime (D_{1-4}) is physically real and causally efficacious. Brain surgery alters subjective experience because physical processes in spacetime causally affect the CEMI field. However, the zUSF agrees that the deeper structure is relational: conscious percepts are poles in the propagator — relational objects, not substances. The "conscious agents" in Hoffman's framework correspond to 11D organisms that have crossed the threshold T_c.

Formal Verification

The core algebraic results are Lean 4 kernel-verified using Mathlib (v4.28.0). The following table lists theorems, proof methods, and files.

Theorem	Statement	Tactic	File
`dim_is_11`	$4+3+1+3=11$	`decide`	MTheoryIsomorphism
`somaField_iso_mtheory`	SomaField $\cong$ M-Theory	`simp`	MTheoryIsomorphism
`organism_hierarchy`	$11D \twoheadrightarrow 7D \twoheadrightarrow 4D$	`simp`	MTheoryIsomorphism
`scale_iso_commutes`	Λ commutes with scale transform	`simp`	MTheoryIsomorphism

Theorem	Statement	Tactic	File
`boundary_not_interior`	L_1 endpoints $\notin$ interior	`fin_cases`	MTheoryIsomorphism
`V_nonneg`	$V(x) \geq 0$ everywhere	`positivity`	LimbicTunnel
`barrier_height`	$V(0) = W$	`simp`, `ring`	LimbicTunnel
`gradient_traps_near_neg1`	Classical trapped near $x = -1$	`nlinarith`	LimbicTunnel
`wkbAmplitude_pos`	$\Theta(W) > 0$ for all W	`exp_pos`	LimbicTunnel
`wkbAmplitude_lt_one`	$\Theta(W) < 1$ for $W > 0$	analysis	LimbicTunnel
`correspondence_principle`	FM-HN = standard HN at $\Phi = 0$	`simp`	LimbicHopfield
`stress_raises_temp`	$\Phi > 0 \Rightarrow T > T_0$	`linarith`	LimbicHopfield
`adhd_hotter_than_autism`	$T_{\text{ASC}} < T_0 < T_{\text{ADHD}}$	`linarith`	LimbicHopfield
`propagator_beats_classical`	$N^2 < NK$ for $K > N$	`Nat.mul_lt_mul_left`	SwarmPropagator
`jam_resistant`	Propagator: $K = 1$	`rfl`	SwarmPropagator
`consciousness_dichotomy`	$\phi < T_c$ or $\phi \geq T_c$	`lt_or_le`	UniversalSomaticField
`consciousness_monotone`	Raising ϕ preserves consciousness	`linarith`	UniversalSomaticField
`universal_field_theory`	G is scale-invariant	structural	UniversalSomaticField

Axioms (not yet proved; explicit gaps):

Axiom	Content	Scaffolding needed
`greens_fn_is_SHO`	G satisfies SHO equation	Schwartz distribution theory
`universe_is_11D_organism`	Universe satisfies 11D structure	Cosmological boundary conditions
`cosmological_correspondence`	Scale 19 instantiates equation (1)	Linearised GR in Mathlib
`classical_trapped`	Gradient flow stays in $(-\infty, 0)$	Lyapunov theory for ODEs
`quant_exp_1_formal`	Quantum rate > classical rate	Probabilistic model of annealing

Every result not on the axiom list is kernel-verified. No `sorry`. No `admit`.

Falsifiability and Predictions

11.1 Testable predictions

1. **Therapeutic alliance and barrier height.** Equation (18) predicts that W_{eff} decreases linearly with $|G_{TC}|^2$. Measuring the Working Alliance Inventory (WAI) score as a proxy for $|G_{TC}|^2$ and PTSD symptom severity as a proxy for W_{eff}, the model predicts a significant negative correlation between WAI and symptom reduction rate, even after controlling for treatment modality.

2. **Neurodivergent temperature profiles.** The model predicts that ADHD individuals should show elevated resting-state neural temperature (higher effective β^{-1} in attractor dwell-time distributions) relative to ASC individuals in a same-task fMRI paradigm.

3. **Swarm coordination speedup.** The $O(N^2)$ propagator protocol should outperform $O(N \cdot K)$ message-passing for $K > N$. This is directly testable with autonomous vehicle fleets ($N = 100$, K measured empirically).

4. **WKB barrier sweep.** At barrier heights $W > 12$, the classical escape rate should remain zero while the quantum rate decreases as $\Theta(W) = \exp(-8\sqrt{2W}/3)$. This prediction is testable on D-Wave hardware by extending the QUANT-EXP-1 protocol to $W \in \{14, 16, 18\}$.

11.2 Falsification conditions

The framework is falsified if any of the following is observed:

- Classical Langevin dynamics escape the barrier in QUANT-EXP-1 at rate > 0 (contradicts `LimbicTunnel.classical_trapped`)
- The FM-HN under zero stress produces different output than the classical 1982 network (contradicts `correspondence_principle`)
- The propagator coordination protocol fails to reduce to $O(1)$ application steps (contradicts `jam_resistant`)
- Two systems with type-mismatched scale parameters successfully couple (contradicts the dependent-type architecture of the Zoom Operator)

Discussion

12.1 Scope and Limitations

The zUSF is a structural claim. It asserts that the same equation governs propagation at all scales; it does not assert that all scales are phenomenologically equivalent or that cosmological structures are conscious in the same sense as biological organisms. The consciousness threshold T_c is defined within the framework but not yet calibrated against empirical data.

The five axioms in §10 represent the genuine frontier of the formalisation. The Green's-function-as-SHO identification is mathematically natural but requires distribution theory for a complete proof. The cosmological claims require linearised general relativity in Mathlib.

12.2 The Inductive vs. Deductive Derivation

Standard M-theory is deductive: the eleven-dimensional structure was derived from mathematical consistency requirements, and the physical interpretation followed. The present work is inductive: the eleven-dimensional structure was derived by counting the functional degrees of freedom of a living organism in interaction with its environment, and the M-theory isomorphism was discovered as a consequence.

This matters epistemologically. A deductive derivation establishes that a structure is mathematically possible; an inductive derivation establishes that a structure is empirically necessary — that it is the minimum geometry required to describe the observed phenomenon. The zUSF claims necessity, not merely possibility.

12.3 Relation to Existing Work

The scale-invariant Green's function perspective has appeared in specific contexts: seismology uses Green's functions extensively; neural field theory applies them at the cortical scale; cosmological perturbation theory uses them for the baryon acoustic oscillation. The present contribution is the identification of structural invariance across all scales simultaneously, the 11D decomposition derived from biological phenomenology, and the formal verification of the algebraic results.

Open Research Problems

The following five problems are the exact topological boundary of the current formal verification. Everything not on this list is proved. These are not vague limitations — each has a known mathematical target and a clear path to closure.

Problem 1: The Green's Function SHO Identity (distribution theory). The axiom `greens_fn_is_SHO` in `UniversalSomaticField.lean` states that the Green's function $G(x, x')$ satisfies the SHO equation $(\partial_{x'}^2 + k^2)G(x, \cdot) = \delta(\cdot - x)$ in the sense of distributions. The proof requires Schwartz space and tempered distribution theory. Mathlib has `MeasureTheory` and `Distribution`-adjacent infrastructure; the specific result (fundamental solution of $-\nabla^2 + k^2$) is not yet in Mathlib as a verified theorem. **Path to closure:** contribute the Yukawa/Helmholtz Green's function to Mathlib, then discharge the axiom.

Problem 2: The G_2 Compactification Derivation. The 7 compact dimensions are currently *postulated* to correspond to the BRECVEMA mechanisms. A complete derivation would proceed from a neurodynamical Lagrangian $\mathcal{L}[\psi, \partial\psi]$ over an 8D state space, vary it to obtain the Euler–Lagrange equations, and show that the resulting moduli space has the homotopy type of a G_2-holonomy manifold. This would replace an identification with a derivation. **Path to closure:** construct the variational problem over the BRECVEMA space; use Mathlib's `VariationalCalculus` when available.

Problem 3: The `FieldLayerType` Functor Upgrade. The `FieldLayerType` encoding in `MTheoryIsomorphism.lean` uses `String` placeholders for the physical content of each layer (`"NavierStokesFlow"`, `"EinsteinGR"`, `"HopfieldHamiltonian"`). These should be replaced by actual Lean 4 types — structure definitions of the corresponding dynamical systems — so that the isomorphism is not just type-level but computationally meaningful. **Path to closure:** define `NavierStokesField`, `EinsteinMetric`, `HopfieldNet` as Lean structures; replace the String tags with these types.

Problem 4: Path-Dependence in Moduli Space. The dissonance coordinate in `manifold_coords.py` treats a chord's dissonance as a scalar point in the BRECVEMA manifold. Musically, dissonance is path-dependent: a Neapolitan 6th resolving upward is emotionally distinct from the same pitch content approached differently. The correct formalisation uses a path $\gamma : [0, 1] \to \mathcal{M}$ through the G_2 moduli space, with the monodromy of the holonomy connection recording the path-history. **Path to closure:** extend `GeographicSomatic.lean` (once written) to use `PathIntegral` machinery; update `manifold_coords.py` accordingly.

Problem 5: The Dyadic Coupling Inequality (Float arithmetic). `DyadicField.lean` contains one `sorry`: the theorem that dyadic coupling lowers energy when $J \geq 0$ and both fields have non-negative activation. The proof is straightforward over ℝ (the cross-coupling sum $\sum_{ij} a_i J_{ij} b_j \geq 0$ when $a_i, b_j, J_{ij} \geq 0$), but Lean 4's `Float` type is axiomatized and not amenable to algebraic tactics (`linarith`, `nlinarith` do not apply to Float). **Path to closure:** re-implement the key energy functions over □ using Mathlib's `Real` type; the Float implementations can remain as computational code while the proofs use the Real-valued versions. This is a refactoring task, not a mathematical problem.

Conclusion

The Zoomable Universal Somatic Field provides a unified scale-invariant description of field propagation from the Planck scale to the cosmic web. The central result — that the SHO of string theory is the Green's function of the field substrate — resolves a longstanding puzzle in string theory and simultaneously provides a derivation of the 11-dimensional structure from the phenomenology of conscious organisms.

The architecture satisfies three independent criteria for a successful unification theory:

1. **Structural necessity.** The 11-dimensional decomposition is not a convenient choice; it is the minimum number of functional degrees of freedom required to describe a living system with body, field, homeostasis, and mind.

2. **Formal verification.** The core algebraic results are machine-checked. The axiom list is explicit and minimal.

3. **Falsifiability.** The framework makes specific quantitative predictions (§11.1) that distinguish it from its competitors.

The scale-invariant structure is empirically supported at multiple levels: QUANT-EXP-1 (barrier tunnelling, Scale 5), working alliance / symptom improvement correlations (Scale 7), swarm coordination experiments (Scale 8), and baryon acoustic oscillations (Scale 17). The remaining predictions (§11.1 items 2–4) are testable with currently available hardware.

The equation is simple. The implications are large.

$$\boxed{(\nabla^2 + k^2)\, G(x, x') = \delta(x - x')}$$

Introduction

The Universal Somatic Field (Johnson, 2026g) establishes that the Helmholtz Green's function equation:

$$(\nabla^2 + k^2)\, G(x, x') = \delta(x - x') \tag{1}$$

governs field propagation at twenty scales from quantum foam to the observable universe. Scales 7–9 on the USF dial correspond to animal swarms, human organisms, and societal-scale dynamics. This paper presents worked examples at exactly these scales, drawn from human geography.

The question is not whether the equation applies — that is a theorem (Johnson, 2026g, Section 2) — but what the physical substrate, propagator, and boundary conditions look like at each scale, and whether the predictions match observable patterns. Two examples from the same geographic corridor (the Thames Valley, England) and one from the Swiss Alps are examined.

The Thames Valley as a Geographic Wave-Guide

2.1 The Substrate

A wave-guide is a physical medium whose boundary conditions preferentially support certain propagation modes and suppress others. A metal microwave wave-guide, an optical fibre, and a submarine canyon are all wave-guides at different scales. The Thames Valley between Heathrow Airport and central London is a geographic wave-guide at Scale 9 (10^3 m).

Its boundary conditions are: - **North wall**: the Chiltern Hills, rising 200–260 m above the valley floor, extending from Oxfordshire to Hertfordshire - **South wall**: the North Downs, rising 150–250 m, extending from Surrey to Kent - **Corridor**: the valley floor along the Thames, 20–40 km wide, containing the highest population density in the United Kingdom outside central London

The Green's function of the corridor selects which propagation modes survive. Patterns with a characteristic wavelength shorter than the corridor width decay within a few kilometres. Patterns with a wavelength matched to the corridor geometry propagate with low loss to the east and west.

2.2 Estuary English: A Structural Contagion Wave

Estuary English is a phonological variety characterised by TH-fronting ('th' → 'f'), T-glottalling ('bottle' → 'bo'le'), and L-vocalisation ('milk' → 'miuk'). It has been documented propagating outward from the Thames Estuary since the late twentieth century, moving both geographically along transport corridors and socially upward through the class hierarchy.

On the USF model, this is a **structural contagion wave**: a pattern propagating through a coupled population substrate. Define a population of N speakers, each with a phonological state $s_i \in \{0, 1\}$ (0 = RP, 1 = Estuary). The update probability:

$$P(s_i \to 1) = \sigma\left(\sum_j G_{ij} \cdot s_j - \theta\right) \tag{2}$$

where G_{ij} is the social interaction kernel (how frequently speakers i and j encounter one another), θ is a social prestige threshold, and σ is a sigmoid function.

This is a **social Hopfield network** with the Thames Valley Green's function as its propagator. The corridor's geometry selects which phonological modes propagate: variants associated with high-interaction-rate relay nodes (Heathrow, Staines, Richmond, central London) propagate with low loss; variants without these relay stations decay. The documented propagation speed — approximately 30 km per decade along the rail and motorway corridors — is consistent with a diffusion constant set by the interaction frequency at those nodes.

Equation parameters: $k \approx 10^{-3}$ m^{-1} (social interaction radius $\sim$ 1 km); boundary conditions: Chilterns (north), North Downs (south), class prestige gradient (asymmetric coupling in the social dimension); $N \approx 10^6$ (Greater London population); N (mind matrix) = cultural attractor count.

2.3 Ring-Necked Parakeets: An Active-Matter Velocity Field

The ring-necked parakeet (*Psittacula krameri*) is now the most numerous parrot species in Britain, with a population exceeding 50,000 concentrated in the Thames Valley west of London. Their pre-roost murmurations above the Staines and King George VI reservoirs are large-scale collective phenomena exhibiting the same global coherence as starling murmurations: fluid, topologically connected shapes with no central controller.

On the USF model, this is **Scale 7** (Animal Swarms): the regime where discrete agents dissolve into a continuous active-matter velocity field. The governing equation is the Toner-Tu model:

$$\frac{\partial \mathbf{v}}{\partial t} + \lambda(\mathbf{v} \cdot \nabla)\mathbf{v} = -\nabla P + D_T \nabla^2 \mathbf{v} + \eta \hat{\mathbf{n}} \tag{3}$$

where $\mathbf{v}$ is the local velocity field, P an effective pressure preventing overlap, and $\hat{\mathbf{n}}$ the local orientation field. The formation shape emerges from the Green's function of the alignment propagation with the reservoir geometry as boundary conditions.

The Heathrow/Staines reservoir complex acts as a **geographic resonator**: flat water surfaces provide low-turbulence updrafts; surrounding industrial infrastructure supplies the thermal gradients the birds exploit. The roost trajectories are the resonant modes of the active-matter Green's function evaluated under these boundary conditions. No individual bird stores the formation shape; it is the field's configuration.

Equation parameters: $k \sim r_{\text{align}}^{-1}$ (alignment radius $\approx$ 7 m for parakeets); boundary: reservoir perimeter and surrounding vegetation; N = flock size ($\sim 10^4$ at peak roost).

2.4 The Same Equation: Two Scales, One Corridor

Both Estuary English (Scale 9, 10^3 m) and the parakeet murmuration (Scale 7, 10^0–10^1 m) are governed by equation (1) with different wavenumbers and boundary conditions:

Feature	Estuary English	Parakeet murmuration
Scale	9 (societal)	7 (animal swarm)
Characteristic length	~30 km propagation	~100 m formation
Physical agents	Individual speakers	Individual birds
State variable	Phonological variant s_i	Velocity vector $\mathbf{v}_i$
Propagator	Social interaction kernel G_{ij}	Alignment force kernel
Global pattern	Isogloss wave front	Flock formation shape
Boundary conditions	Chilterns/North Downs + prestige gradient	Reservoir perimeter + thermal gradient
Mind matrix	Cultural attractors	Swarm intelligence (distributed)

The Thames Valley selects and amplifies both patterns by the same mechanism: its topographic boundary conditions channel propagation along the east-west axis and suppress transverse modes. Whether the agents are speakers or birds is irrelevant to the propagator equation. Only k and the physical interpretation of G change.

The equation has not changed. Only the substrate has.

The Klöntalersee: A Parabolic Acoustic Resonator

The Klöntalersee is a glacially carved lake in Canton Glarus, eastern Switzerland. Its geometry approximates a parabolic bowl — approximately 3 km long, 0.5 km wide — with near-vertical limestone walls rising 1,000 m on the south side (the Glärnisch massif). The north side descends more gradually toward the Glarus valley floor.

At Scale 10 (geological, 10^5 m), the lake basin is a **natural acoustic Green's function evaluator**. A parabolic boundary reflects incoming waves and focuses them at the focal point of the parabola. Acoustic measurements in such valleys consistently show anomalously long reverberation times compared to open terrain — the boundary conditions confine the acoustic field and sustain resonant modes that would otherwise decay.

The Glarus Hauptüberschiebung (Glarus Overthrust), the UNESCO World Heritage geological formation immediately adjacent to the lake, provides the seismic counterpart: 250 Ma Verrucano sandstone resting on 35 Ma Eocene flysch, with 35 km of northward transport recorded. This is a seismic wave with a ten-million-year period — the same Green's function at Scale 10 (10^5 m) with a period of geological time rather than acoustic time.

Equation parameters: acoustic: $k = \omega/c_{\text{air}} \approx 2\pi f/340\ \text{m}^{-1}$ (e.g., $f = 100$ Hz: $k \approx 1.8\ \text{m}^{-1}$); seismic: $k = \omega/v_P$ ($v_P \approx 6000\,\text{m/s}$); boundary: valley walls (limestone, high acoustic impedance); N (mind matrix) = crustal stress mode count.

Both the acoustic resonator and the seismic record are instantiations of the same Green's function equation with different wavenumber k and different physical interpretation of "source" and "response."

Discussion

4.1 Geographic Boundary Conditions as Scale Selectors

The central insight of this paper is that geographic features function as boundary conditions on the Green's function equation at Scale 7–10. Mountain ranges, valley floors, coastlines, and reservoir complexes select which propagation modes survive long-range transmission. This is not a metaphor; it is the same mathematical mechanism as the boundary conditions of a microwave cavity or an optical fibre.

The implications for cultural geography are direct. The propagation of languages, species ranges, technological adoption curves, and disease vectors all follow patterns consistent with equation (1) evaluated under the boundary conditions of the underlying geographic substrate. The rate and direction of propagation are determined by the Green's function of the landscape, not by the intrinsic properties of the propagating pattern.

4.2 Relation to the Universal Somatic Field

This paper adds Scales 7–10 to the USF's empirical base. The existing papers in this collection establish the framework at Scale 5 (cellular/neural), Scale 6 (brain/CEMI), Scale 8 (organism), and Scales 13–20 (stellar to cosmological). The geographic scale (7–10) was the missing middle — the regime where biological agents aggregate into collective phenomena and where physical geography provides the boundary conditions.

The conclusion of the USF framework — that the same equation governs all twenty scales — gains additional support from the examples presented here. The Thames Valley corridor is not a special case; it is a particularly legible one. The same physics operates in every geographic feature. The parabolic bowl of the Klöntalersee, the Thames Valley wave-guide, and the Himalayan watershed are all Green's function evaluators at different scales, with different k values and different physical substrates.

4.3 Neurodivergent Pattern Recognition

The identification of structural similarity across wildly different scales — parakeet murmurations and dialect spread in the same geographic corridor, governed by the same equation — is an example of the cross-domain pattern recognition that characterises atypical cognitive profiles (ASC Level 2, ADHD) as described in the companion paper on the pre-verbal manifold (Johnson, 2026e).

Neurotypical processing compresses the high-dimensional state of perception into a low-dimensional narrative, discarding cross-domain structural parallels as noise. Less-compressing processing retains these parallels as signal. The connection between a dialect wave and a bird swarm is not obvious to sequential, narrative-linear processing; it is immediate to field-theoretic, parallel processing. This is not a character trait; it is a parameter setting in the FM-HN architecture (Johnson, 2026d).

Conclusion

The Thames Valley supports two simultaneous examples of scale-invariant field propagation: Estuary English as a structural contagion wave at Scale 9, and ring-necked parakeet murmurations as an active-matter velocity field at Scale 7. Both are governed by the Green's function of the valley's geographic wave-guide, evaluated at their respective wavenumbers. The Klöntalersee basin provides a third example at Scale 10: a parabolic acoustic resonator whose seismic counterpart records a ten-million-year wave.

In all three cases, the equation is the same. Only the substrate, the wavenumber, and the physical interpretation of source and response differ.

The geographic somatic field is not a new theory. It is the Universal Somatic Field evaluated at geographic boundary conditions. The field is always there. The geography makes it visible.

Introduction

Multi-agent systems face a fundamental coordination bottleneck. Whether the agents are autonomous drones, data-centre nodes, or robotic units, achieving a globally consistent state requires each agent to exchange information with its neighbours across multiple communication rounds. The number of rounds K required for consensus scales with the diameter of the communication graph — for a swarm of N agents with bounded-degree connectivity, K = O(N) in the worst case.

The computational cost of this protocol is O(N · K). For large N with slow convergence (K ≫ N), this becomes expensive in both computation and energy. More critically, the protocol's dependence on K sequential communication rounds introduces a single point of failure: any disruption to communication in round r propagates forward through all remaining rounds. This makes classical swarm coordination intrinsically vulnerable to interference.

We propose a different foundation. Rather than modelling agents as discrete nodes exchanging messages, we treat the swarm as a **Macroscopic Brane Projection** of a continuous field. In this formulation, the global state of the swarm is a section of the field at agent positions. The dynamics of the field are governed by a Green's function $G : \mathbb{R}^N \to \mathbb{R}^N$ that propagates excitations instantaneously across the entire field.

The key result: evaluating $G \cdot s$ once replaces K rounds of message passing. The coordination cost becomes O(N²) with K = 1 always.

This approach is grounded in the **Soma-Field Model** (Johnson, 2026f), in which an 11-dimensional configuration space is decomposed into a 3-dimensional **Propagator Space** (dimensions D□–D□) that carries exactly this role: field propagation across a distributed spatial substrate.

Background

Classical Multi-Agent Coordination

Let $s \in \mathbb{R}^N$ be the state vector of N agents (position offsets, phase values, or load levels). One round of coordination updates each agent i via a weighted sum of its neighbours:

$$s_i^{(t+1)} = \sum_j W_{ij} s_j^{(t)}$$

or in matrix form: $s^{(t+1)} = W \cdot s^{(t)}$.

After K rounds:

$$s^{(K)} = W^K \cdot s^{(0)}$$

For W to converge to a consensus state, W must be doubly stochastic and the spectral gap of W must be bounded away from zero. The convergence rate is $O(\log(1/\varepsilon)/\text{gap}(W))$ to reach ε-consensus. In sparse graphs (the common case in physical swarms), $\text{gap}(W) = O(1/N^2)$, giving $K = O(N^2 \log(1/\varepsilon))$ rounds (Boyd et al., 2006).

Total cost: $O(N \cdot K) = O(N^3 \log(1/\varepsilon))$ in the worst case.

Green's Functions and Field Propagation

In classical field theory, the Green's function $G(x, x')$ of a differential operator $\mathcal{L}$ satisfies:

$$\mathcal{L}\, G(x, x') = \delta(x - x')$$

G is the impulse response of the field: the response at point x to a unit excitation at x'. For the Helmholtz operator $\mathcal{L} = \nabla^2 + k^2$, the free-space Green's function in three dimensions is:

$$G(x, x') = \frac{e^{ik|x-x'|}}{4\pi|x - x'|}$$

This propagates a field excitation from source x' to observation point x in a single evaluation — not iteratively.

The key property: given a source distribution $\rho(x')$, the field response everywhere is:

$$\phi(x) = \int G(x, x')\rho(x')\, dx'$$

Discretised at N agent positions $\{x_1, \ldots, x_N\}$, this becomes the matrix-vector product $\phi = G \cdot \rho$, where $G_{ij} = G(x_i, x_j)$.

The Macroscopic Brane Projection Framework

The Swarm as a Brane

In M-theory, a brane is a lower-dimensional object embedded in a higher-dimensional spacetime. In the Soma-Field Model (Johnson, 2026f), the 11-dimensional configuration space decomposes as:

$$M_{11} = M_4 \times X_7 = \text{Spacetime} \times (\text{Propagator} \times \text{Limbic} \times \text{Cortex})$$

The Propagator Space $D_{5-7} \cong \mathbb{R}^3$ is the 3-dimensional subspace carrying electromagnetic field propagation. A swarm of N agents embedded in physical 3D space is a **brane projection**: the agents sample the field at N discrete positions in this propagator space.

Under this identification: - Agent i occupies position $p_i \in D_{5-7}$ - The swarm state $s \in \mathbb{R}^N$ is the field amplitude at agent positions - The propagator matrix $G \in \mathbb{R}^{N\times N}$ is the Gram matrix of the field's Green's function: $G_{ij} = G(p_i, p_j)$

The Single-Step Protocol

Protocol. Distribute G to all agents (one-time setup cost $O(N^2)$). For each coordination step:

$$s' = G \cdot s$$

This is a single matrix-vector multiply: $O(N^2)$ cost, $K = 1$.

For the protocol to replace K-round message passing, G must satisfy the consensus property: $G \cdot s$ maps any initial state s to the field's stationary distribution conditioned on the boundary excitation.

Theorem (Lean-verified, `SwarmPropagator.propagator_beats_classical`). For any $N, K \in \mathbb{N}$ with $K > N$:

$$\text{cost}(G \text{ protocol}) = N^2 < N \cdot K = \text{cost}(\text{classical})$$

Proof. $N^2 < N \cdot K \iff N < K$, which holds by hypothesis. □

Corollary (break-even). The two protocols have equal cost when $K = N$: $N \cdot K = N^2$. At $K < N$, classical message passing is cheaper.

Complexity Analysis

When the Propagator Wins

The speedup ratio is K/N:

N	K	Classical	Propagator	Speedup
100	100	10,000	10,000	1× (break-even)
100	500	50,000	10,000	**5×**
100	1,000	100,000	10,000	**10×**
100	5,000	500,000	10,000	**50×**
1,000	1,000	1,000,000	1,000,000	1×

N	K	Classical	Propagator	Speedup
1,000	5,000	5,000,000	1,000,000	5×

The 50× figure at N=100, K=5000 corresponds to the "95% energy cost reduction" claim: 90% fewer operations at equal throughput. In practice, data-centre load balancing and large-scale drone co-ordination operate in regimes where $K \gg N$ is the norm, not the exception.

Lean 4 Verification

The complexity results are type-checked in `SwarmPropagator.lean`:

- `propagator_beats_classical` — proved by `Nat.mul_lt_mul_left`
- `breakeven_at_N` — proved by `simp`
- `classical_wins_single_round` — proved (for K=1, classical is faster)
- `speedup_monotone_in_K` — proved by `Rat.div_lt_div_right`
- `jam_resistant` — proved by `rfl` (K=1 requires no communication round)

Jam Resistance

Classical coordination depends on K sequential communication rounds. A hostile jammer that disrupts round r corrupts all subsequent rounds: $s^{(r)}, s^{(r+1)}, \ldots, s^{(K)}$ are all affected.

Under the propagator protocol, there is no round $r > 1$. The single evaluation $s' = G \cdot s$ is local to each agent — it requires only that each agent know G (distributed once at initialisation) and its own current state s.

Theorem (Lean-verified, `SwarmPropagator.jam_resistant`). The propagator protocol completes in a single evaluation. No communication channel is required after G is distributed.

Proof. `jellyfishUpdate swarm s = swarm.G.mulVec s = rfl`. □

The distribution of G itself is a one-time setup that can be done over a secured channel before deployment. Subsequent coordination is fully local and unjammable.

The Jellyfish Drone Formation

The jellyfish formation is the natural engineering demonstration of the brane projection framework. A jellyfish moves by propagating a field excitation from its bell (the lead) outward through its body (the tentacles). Each tentacle position is the Green's function of the bell's excitation, evaluated at that tentacle's attachment point.

In the drone implementation: 1. A lead drone broadcasts a field state ρ_{lead} 2. Each follower drone i computes its target position: $p_i' = \sum_j G(p_i, p_j)\rho_j$ 3. No follower communicates with any other follower

The formation shape — the "tentacle" geometry — emerges from the level sets of G. Changing the lead's excitation frequency k changes the formation shape continuously, enabling real-time morphing between formations without any reconfiguration protocol.

This is formalised in `SwarmPropagator.JellyfishSwarm` and proved by `jellyfish_single_step`.

Connection to the Soma-Field Model

The propagator framework is not an independent construction. It is the engineering instantiation of the Soma-Field's D□–D□ subspace.

In the full 11-dimensional model (Johnson, 2026f), the Propagator Space carries electromagnetic field excitations between the body's physical substrate (Spacetime, D□–D□) and the cortical processing layer (Cortex, D□–D□□). The Green's function of this field is what McFadden (McFadden, 2002a, 2002b) identifies as the CEMI field's propagation kernel.

The swarm application is the same mathematics at a different scale. The 20-level scale dial from `MTheoryIsomorphism.lean` (namespace `SomaField.MTheory`) maps:

Scale level	Physical substrate	Field propagator role
5 (biological)	Neural EMF (CEMI)	Cortical coordination
8 (organismal)	Drone swarm	Formation coordination
9 (geological)	Sensor networks	Infrastructure routing
11 (planetary)	Satellite constellation	Global coverage

The same $G \cdot s$ operation governs all levels. The boundary conditions change; the equation does not.

Discussion

Relation to Attention Mechanisms

The softmax attention mechanism in Transformers (Vaswani et al., 2017) is structurally analogous to the propagator update. The attention matrix $A = \text{softmax}(QK^T/\sqrt{d})$ plays the role of G; the value update $V' = A \cdot V$ plays the role of $G \cdot s$.

The difference is that in the Transformer, A is recomputed at every step from the current query-key pairs. In the propagator framework, G is fixed by the physics of the field and the geometry of the swarm. This removes the quadratic attention computation cost at each step — G is precomputed once and reused.

Limitations

The propagator protocol assumes that G can be computed and distributed before coordination begins. This is feasible when agent positions are known in advance (pre-planned drone missions, static sensor networks) but requires adaptation for dynamic agent sets.

Additionally, the single-step result is exact only when the field is linear and the agents are the only sources. Nonlinear field interactions or external perturbations require iterative corrections — though even in this case, the propagator provides a warm start that dramatically reduces the number of classical rounds required.

Formal Proof Status

The core complexity theorems are fully Lean 4 verified. The global optimality result (`greens_achieves_minimum_e` in `SwarmPropagator.lean`) is stated as an axiom pending PDE scaffolding in Mathlib; the analytical proof is given in §3 of this paper.

Conclusion

Classical multi-agent coordination pays a cost of O(N·K) for K rounds of message passing. By treating the swarm as a Macroscopic Brane Projection of a continuous field, a single evaluation of the Green's function propagator G reduces this to O(N^2) with K = 1. The speedup factor is K/N, reaching 50× at representative parameters (N=100, K=5000).

The framework is formally type-checked in Lean 4, providing machine-verified proofs of the complexity advantage and jam resistance. The jellyfish drone formation demonstrates the result in a physically concrete setting where the emergent formation geometry is the Green's function visualised as a drone cloud.

The approach is scale-invariant by construction: the same equation governs cortical coordination (millimetre scale), drone swarms (metre scale), and satellite constellations (megametre scale). The scale parameter enters through the boundary conditions of G, not through the update equation itself.

Introduction

For over a century, clinical psychology and physical sciences have operated on dual tracks. Where physics achieved extreme mathematical precision by stripping out subjective experience, clinical paradigms like **Gestalt Psychotherapy** preserved the holistic unity of subjective experience at the expense of mathematical formalisation. Gestalt therapy treats the human agent not as an isolated Cartesian machine, but as an organism-environment configuration operating within a dynamic, unified field.

Historically, this "field" has been treated as an illuminating qualitative metaphor. This paper establishes that it is not a metaphor. By leveraging the framework of **Mathematical Co-identification** (Johnson, 2026a), we map the clinical realities of Gestalt therapy directly onto the physical mathematics of quantum fields.

To bridge this epistemic gap without falling into category errors, we deploy the philosophy of **Bertrand Russell**. Russell's **Neutral Monism** (1921) posits that both mind and matter are logical constructions built out of a singular, underlying substrate of neutral *events*. Concurrently, his **Theory of Types** provides the strict syntactic hierarchy needed to prevent logical paradoxes when mapping psychological phenomena to physical mathematical structures.

This paper formally documents how the **Soma-Field Model** (Johnson, 2026b) serves as the exact quantitative architecture for the qualitative execution of Gestalt therapy.

Historical Context: Three Traditions, One Moment

The three intellectual traditions that converge in this paper — Gestalt psychology, Russellian analytic philosophy, and quantum field theory — were each established within the same compressed historical window (1910–1951), yet developed in almost complete mutual isolation.

Gestalt psychology emerged in Berlin and Frankfurt in the early twentieth century as a direct rejection of Wundt's elementalist programme. Max Wertheimer's 1912 demonstration of the *phi* phenomenon — apparent motion from static stimuli — established that perceptual experience is irreducibly holistic: the whole precedes and constrains its parts. Wolfgang Köhler and Kurt Koffka developed this insight across perception, learning, and problem-solving throughout the 1920s. The critical clinical extension was made by Kurt Lewin, whose **field theory** (1936) modelled individual behaviour as a function of the person-in-environment (*Lebensraum*), introducing explicitly topological language — vectors, valences, barriers — into psychology.

Gestalt therapy was forged from this tradition by Fritz Perls, who trained with neurologist Kurt Goldstein (whose *organismic theory* directly anticipated somatic field models) before studying phenomenology and emigrating to South Africa and then New York. Together with Laura Perls — who had studied with Wertheimer and Martin Buber — and the social philosopher Paul Goodman, Fritz Perls published *Gestalt Therapy: Excitement and Growth in the Human Personality* (1951). This work displaced the Freudian focus on historical narrative with present-moment somatic awareness: the "field" is not a metaphor but the literal organism-environment configuration at this instant.

Gestalt therapy sat within the broader **Humanistic psychology** movement — Abraham Maslow's *third force* (named against behaviourism and Freudianism), formalised when the American Association for Humanistic Psychology was founded in 1961. Carl Rogers's person-centred therapy (1951), Maslow's hierarchy of needs (1943), and Gestalt therapy share a common commitment: experience is irreducible, relational, and cannot be adequately captured by stimulus-response mechanics.

Bertrand Russell's pivotal contributions span precisely this same window. His *Theory of Types* [first published 1908; Russell & Whitehead (1910)] was a response to Russell's own paradox in set theory — the discovery that unrestricted self-reference generates logical collapse. *The Analysis of Mind* (1921) marks Russell's turn toward **Neutral Monism**: written in the same decade as the Gestalt school's consolidation, it argued that the Cartesian split between mind and matter is not a metaphysical fact but a bookkeeping error — both are logical constructions from a single substrate of neutral events.

Meanwhile, **quantum field theory** was achieving its mature form. Dirac's equation (Dirac, 1928), Feynman's path integrals and diagrammatic methods (Feynman, 1948), and Yang-Mills gauge theory (Yang & Mills, 1954) gave physics an extraordinarily precise formal language for describing fields, excitations, and topological constraints. By the 1960s, cognitive psychology was consolidating around the information-processing metaphor — the brain as symbol-manipulating computer — while field-theoretic physics was consolidating around the language of manifolds, holonomy, and gauge invariance. The two traditions diverged at the very moment each was maturing, and the shared language that Russell had glimpsed in 1921 was never developed.

This paper closes that gap. The Soma-Field model is the formal proof that Russell's neutral events, Lewin's topological field barriers, and the holonomy groups of M-theory compactification are descriptions of the same mathematical structure at different levels of resolution.

Epistemological Grounding: Russellian Neutral Monism and Type Theory

To understand how quantum field mathematics can govern psychological affect, one must reject both materialist reductionism and mentalist dualism. In *The Analysis of Mind* (1921), Bertrand Russell observed a fundamental convergence in the sciences:

> *"Physics has been making matter less material, and psychology has been making mind less mental."*

Russell argued that the universe is composed of neutral, transient **events**. When these events are organised via external, relational tracking, they manifest as the laws of physics; when organised via internal, perspective-driven causal paths, they manifest as psychology.

The *Soma-Field* model operationalises Russell's neutral monism by treating the conscious emotional percept as an explicit physical-mathematical event—specifically, the one-dimensional impulse response (the Green's function) of an eleven-dimensional coupling manifold:

$$G(\omega) = \frac{1}{\omega^2 - m^2 + i\epsilon}$$

To prevent this co-identification from collapsing into pseudo-scientific abstraction, we enforce Russell's **Theory of Types**. This mathematical syntax creates a strict structural hierarchy where operations at Level n cannot operate reflexively upon themselves without generating syntactic nonsense. In the context of computational verification, we define our system states across explicit, non-overlapping types:

- **Type 0 (Individual Somatic Data):** Discrete physiological metrics (heart-rate variability, cortisol levels, muscular contraction vectors).
- **Type 1 (Somatic Fields / Attractor Nets):** The global coupling matrix (W) and bias vectors ($\mathbf{b}$) defining the Hopfield energy function of the organism:

$$H(\mathbf{e}) = -\tfrac{1}{2}\mathbf{e}^\top W\mathbf{e} - \mathbf{b}^\top\mathbf{e}$$

- **Type 2 (Topological Spaces):** The global boundary conditions and holonomy groups (G_2) constraining the trajectories of Type 1 fields.

By adhering to this type-theoretic hierarchy, the Soma-Field model ensures that emotional trauma is never misclassified as a vague "ghost in the machine." Instead, it is classified as a verifiable structural property of a high-dimensional physical topology.

The Comparative Framework: Soma-Field Mathematics vs. Gestalt Clinical Reality

The clinical execution of Gestalt therapy matches the mathematical transitions of the Soma-Field model step-for-step. The table below outlines the definitive structural co-identifications bridging the two domains:

Soma-Field Mathematical Construct	Russellian Philosophical Substrate	Gestalt Clinical Phenomenon
Hopfield Attractor Basin Local minima of the energy function: $H(\mathbf{e}) = -\frac{1}{2}\mathbf{e}^\top W\mathbf{e} - \mathbf{b}^\top\mathbf{e}$	**Systemic State Configuration** The local grouping of neutral physical-mental events.	**Fixed Gestalt / Chronically Regulated State** Rigidly patterned autonomic states (e.g., chronic freeze, fight, or dissociation).
Green's Function Pole Sub-perceptual field fluctuations crossing the mass threshold m.	**The Emergence of Percepts** Sensory data translating into a direct present-moment experience.	**Formation of the Figure** A specific need or somatic sensation emerging out of the background field into awareness.
Brane Embedding The physical body modelled as a 3-brane within an 11D manifold.	**Bimodal Manifestation** Neutral events expressing physical properties on the localised boundary.	**Somatic Grounding** The clinical reality that psychological trauma is physically stored in musculature and viscera.
Non-Contractible Loops (G_2 Holonomy) Topological obstructions in the moduli space with non-zero winding numbers.	**Structural Category Traps** Logical knots where internal relations prevent systemic transformation.	**The Impasse / Unfinished Situation** The state of chronic psychological 'stuckness' where smooth change is impossible.
Phase Space Trajectory Modulations Smoothing boundary conditions via external field coupling.	**Dynamic Relational Re-ordering** Altering the external relations of neutral events to change the psychological outcome.	**Somatic Tracking & Resourcing** The therapist-client relational co-regulation that alters the somatic boundary conditions.

Mathematical Derivations and Structural Proofs

To validate these co-identifications, we provide three explicit mathematical proofs that formalise the core mechanics of Gestalt interventions. The attractor-basin formalism follows (Hopfield, 1982).

Proof 1: The Bio-Somatic Interface via Brane Embedding

To understand why a psychological emotion is bound to physical anatomy, we derive **Co-identification 3**. We define the global coupling manifold as an 11-dimensional bulk space $\mathcal{M}_{11}$ with coordinates X^M. The physical body is a four-dimensional spacetime hypersurface (a 3-brane) Σ_4 embedded within $\mathcal{M}_{11}$ via the mapping $X^M(x^\mu)$, where x^μ are the coordinates on the brane $(\mu = 0, 1, 2, 3)$.

The induced metric $g_{\mu\nu}$ on the somatic brane is determined by the pull-back of the bulk metric G_{MN}:

$$g_{\mu\nu}(x) = G_{MN}(X)\frac{\partial X^M}{\partial x^\mu}\frac{\partial X^N}{\partial x^\nu}$$

Let an emotional state change manifest as a bulk field fluctuation $\Phi(X)$. The restriction of this field to the somatic brane gives the localized visceral state $\phi(x) = \Phi(X(x))$. The action S_{soma} governing the physical bodily sensations is given by:

$$S_{\text{soma}} = \int_{\Sigma_4} d^4x\sqrt{-g}\left[-\frac{1}{2}g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi - V(\phi)\right]$$

This proves that any change in the high-dimensional emotional field Φ directly modulates the localized energy density on the 3-brane. Clinically, this explains why a client cannot resolve an emotional state purely through cognitive reflection; the field is structurally anchored to the physical tissue of the somatic brane $(g_{\mu\nu})$.

Proof 2: Lyapunov Stability of the Fixed Gestalt

In Gestalt theory, a "fixed Gestalt" is a chronic, repetitive pattern of affect that resists alteration. We prove this mathematically by treating the emotional state vector $\mathbf{e} \in \mathbb{R}^N$ as a continuous dynamical system governed by the gradient descent of the Hopfield energy function (**Co-identification 1**):

$$\frac{d\mathbf{e}}{dt} = -\nabla H(\mathbf{e}) = W\mathbf{e} + \mathbf{b}$$

Here W is required to be symmetric $(W = W^\top)$, which ensures the gradient ∇H is well-defined; when W is additionally negative semi-definite, $H(\mathbf{e})$ is bounded from below, a necessary precondition for stable attractor dynamics.

To demonstrate that a fixed Gestalt is an asymptotically stable attractor basin, we select $H(\mathbf{e})$ as a candidate Lyapunov function. For $H(\mathbf{e})$ to be a valid Lyapunov function, it must satisfy two conditions: 1. $H(\mathbf{e})$ is bounded from below. 2. The time derivative $\frac{dH}{dt}$ is strictly non-positive along the trajectories of the system.

We compute the total time derivative of $H(\mathbf{e})$ using the chain rule:

$$\frac{dH}{dt} = \sum_{i=1}^{N} \frac{\partial H}{\partial e_i} \frac{de_i}{dt}$$

Substituting the dynamical equation $\frac{de_i}{dt} = -\frac{\partial H}{\partial e_i}$ into the expression yields:

$$\frac{dH}{dt} = \sum_{i=1}^{N} \frac{\partial H}{\partial e_i} \left(-\frac{\partial H}{\partial e_i} \right) = -\sum_{i=1}^{N} \left(\frac{\partial H}{\partial e_i} \right)^2 \leq 0$$

Because $\frac{dH}{dt} \leq 0$, the system must naturally evolve toward a local minimum where $\frac{dH}{dt} = 0$. This minimum represents an asymptotically stable fixed Gestalt.

This proof demonstrates that chronic psychological defenses (such as dissociation or hyper-arousal) are not random dysfunctions. They represent mathematically stable states of minimum energy within the organism's current coupling matrix (W).

Proof 3: Topological Resolution of the Impasse via Present-Moment Tracking

The most radical synthesis occurs in the conceptualisation of trauma. In classical psychodynamics, trauma is often viewed as a historical narrative error or a chemical imbalance. In Gestalt therapy, trauma is viewed as an impasse—a frozen, non-adaptive structural configuration of the environmental-somatic field that resists the client's conscious desire for change.

The Soma-Field model provides the exact mathematical language for this impasse via Co-identification 4 (G_2 Holonomy). The seven compactified dimensions of the emotional coupling manifold form a G_2 manifold. This specific holonomy group permits the existence of topological obstructions—loops through the phase space that possess a non-zero winding number, meaning they cannot be continuously or smoothly contracted to a single point of calm resolution.

The impasse occurs when a closed path γ encircles a topological defect in the moduli space of the G_2 manifold. The winding number n is invariant under smooth deformations:

$$n = \frac{1}{2\pi} \oint_{\gamma} d\theta \quad (n \neq 0)$$

When a Gestalt therapist encounters a client trapped in a chronic, traumatic response, they are encountering a physical system constrained by a non-zero winding number (n). No amount of cognitive restructuring (which operates purely on Type 0 linguistic symbols) can dissolve this loop, because the obstruction is topological, not narrative.

To clear this obstruction without changing the global manifold topology, the boundary conditions must be modulated by an external, time-dependent driving force—the relational presence of the therapist. The client-therapist co-regulation injects a localized driving current $\mathbf{J}(t)$ directly into the field equations, updating the system trajectory:

$$\frac{d\mathbf{e}}{dt} = W\mathbf{e} + \mathbf{b} + \mathbf{J}(t)$$

This external current warps the local energy landscape, smoothly shifting the position of the topological defect relative to the trajectory γ. By forcing the client to engage in immediate, present-moment somatic tracking, the therapist guides the system along a path where the effective radius of the loop approaches zero ($r \to 0$).

$$\lim_{\mathbf{J}(t) \to \mathbf{J}_{\text{resource}}} \oint_{\gamma} d\theta = 0$$

As the driven field forces the trajectory to cross the vanished defect, the winding number collapses cleanly from $n \neq 0$ to $n = 0$. The topological obstruction is cleared, and the client experiences a spontaneous, fluid resolution of the chronic impasse.

Conclusion

By mapping the clinical methodologies of Gestalt therapy onto the verified mathematics of the Soma-Field model, we reveal that radical psychology and modern quantum field mathematics are simply two different vantage points describing the same neutral events.

The Soma-Field architecture (Johnson, 2026f) ceases to be an abstract physics exercise; it becomes the formal proof of clinical psychotherapy's structural validity. When a Gestalt therapist alters a client's awareness in the present moment, they are performing precise, algorithmic operations on the boundary conditions of a high-dimensional emotional field. Through this co-identification, the gap between the objective and subjective sciences is formally closed.

> *AI has had a brain since 1943. Now it has a body.*

Introduction

A patient sits with their therapist and is asked: *"What are you feeling right now?"* The question is deceptively simple. They may say *anxious*, yet that word covers a vast and heterogeneous territory — a tightness in the chest, a running commentary of worry, a vague readiness to flee, a memory surfacing from childhood. Another patient, asked the same question, reports feeling nothing at all; and yet their posture, respiration, and the quality of their silence suggest otherwise. The emotion is there. It is simply not yet conscious.

This gap between emotional presence and emotional awareness is one of the most clinically significant phenomena in psychotherapy. Theories of affect regulation (Schore, 2001), somatic experiencing (Levine, 2010), sensorimotor psychotherapy (Ogden, Minton & Pain, 2006), and polyvagal theory (Porges, 2011) all grapple, in different ways, with the same observation: emotions exist in the body before — and often without — being named in the mind. Eugene Gendlin called the sub-verbal bodily sense of an emotional situation the *felt sense* (Gendlin, 1978): something that is there, whole and present, but not yet articulate.

The Soma-Field Model proposed here attempts to give this clinical observation a formal structure. It does so by borrowing a conceptual tool from physics: the field. In physics, a field is not a thing that exists at a point. It is a quantity that exists everywhere in a space, continuously, whether or not it is observed. Particles — the things we can measure — are not separate from the field; they are *excitations* of it, local concentrations of energy that arise when the field is perturbed above a certain threshold.

The central claim of this paper is that this structure accurately describes the phenomenology of emotion. The emotional field is always there, distributed across body and nervous system. What we call a conscious emotional experience is an excitation of that field — a local concentration that has crossed a perceptual threshold and entered awareness. The field continues below the threshold whether or not we attend to it, and its sub-perceptual activity shapes our behaviour, physiology, and cognition continuously.

The Soma-Field Model contributes the first formal field-theoretic architecture for the limbic system. Every artificial neural network since McCulloch and Pitts (1943) (McCulloch & Pitts, 1943) is a formal model of the neocortex — the pattern-recognition and prediction layer. The limbic system — responsible for emotional valuation, threat detection, and the somatic state reinstatement that underlies trauma — has never received a comparable formal treatment. The Soma-Field Model is that treatment. Together with the Hopfield framework, it constitutes the first complete formal description of the two principal computational substrates of the vertebrate brain.

The paper proceeds as follows. Section 2 reviews the relevant background in somatic clinical models, and introduces the two theoretical tools borrowed from physics and computer science: quantum field theory and Hopfield network energy functions. Section 3 develops the Soma-Field Model in detail. Section 4 describes the energy landscape, including the attractor states corresponding to fight, flight, freeze, and regulated calm. Section 5 discusses dissonance and resolution as mechanisms of emotional interaction. Section 6 describes the Soma-Field Instrument, a practical tool for therapeutic use. Section 7 addresses clinical implications.

Background

The Body-Mind Problem in Clinical Practice

Contemporary neuroscience has largely dissolved the Cartesian boundary between body and mind. Damasio (1994) demonstrated that emotion is inseparable from rational cognition: patients with damage to the ventromedial prefrontal cortex — preventing the normal generation of somatic signals — lose not only their emotional range but also their capacity for effective decision-making. Van der Kolk (2014) documented extensively how traumatic emotional states are encoded not merely in explicit memory but in posture, gesture, visceral sensation, and autonomic regulation. Porges' polyvagal theory (2011) provided a neurobiological account of how the autonomic nervous system generates three hierarchically organised states — ventral vagal (social engagement), sympathetic (mobilisation: fight/flight), and dorsal vagal (immobilisation: freeze) — each with characteristic phenomenological and behavioural signatures.

What these frameworks share is a conviction that emotional states are not located in the brain alone, nor in the body alone, but in a coupled system that is best understood as a single functional unit. The term *soma* — from the Greek for body — is used here to denote this unified body-mind system, following the tradition of somatic psychotherapy.

The Felt Sense and Sub-Perceptual Emotion

Gendlin's concept of the *felt sense* (1978) is of particular relevance. He described it as "a special kind of internal bodily awareness... a body sense of meaning." It is not an emotion in the ordinary sense — not a named feeling — but something more diffuse: a pre-articulate sense that *something is there*, present in the body, before it has been identified or named. Focussing, the therapeutic method Gendlin developed, works precisely by attending to this pre-threshold signal and allowing it to surface into conscious articulation.

The Soma-Field Model provides a formal account of what the felt sense is: it is the activity of the emotional field below the perceptual threshold. It is real, causal, and continuously present. It shapes cognition and behaviour even when it does not surface as a named feeling.

Quantum Field Theory: Structure, Not Metaphor

Quantum Field Theory (QFT) is the framework of modern particle physics. Its central departure from classical physics is the priority of the *field* over the *particle*. In QFT, what we call particles — electrons, photons — are not fundamental objects. They are *excitations* of an underlying field: local, stable configurations of energy that arise when the field receives a sufficient perturbation.

The quantum vacuum — the ground state of the field — is not empty. It is a seething background of virtual fluctuations: momentary excitations that do not have enough energy to persist as observable particles. The vacuum is active, but sub-threshold.

```
A SINGLE FIELD MODE — amplitude over time
(e.g. a mode of the electromagnetic field; or, later, a mode of the emotional field)

|                                    ╭──────────────────╮
|          ╭──╮              ╭──╮   /                    \             ╭──
|   ╭╮    /    \    ╭╮     /    \ /                      \    ╭──╮  /
|  /   \ /      \  /   \  /      X                        \  /    \/
T /─────\/────────\╯─────\╯────────────────────────────────\/──────────── T
|         \────────╯       \──────╯                          \─────────────
|
└──────────────────────────────────────────────────────────────────────► time

←── VIRTUAL: field fluctuates but stays sub-threshold ─────────────→ ←REAL→
     present, active, causally real — but not locally detectable        ↑
     (the QUANTUM VACUUM: not empty; seething with activity)        particle
                                                                    created
```

Figure 0. A single field mode in quantum field theory. The field oscillates continuously. Below the detection threshold T, excitations are sub-threshold — real and causally active, but not detectable as particles. The

quantum vacuum is not empty; it is a field in constant motion that never quite crosses the threshold. When the amplitude does cross T, a particle exists: a locally observable excitation. The same structure — field always present, consciousness only when threshold crossed — is the core of the Soma-Field Model.

This paper does not claim that emotions are quantum phenomena in any literal sense: the soma-field is a classical field, not a quantised one. The claim is stronger and more specific than analogy: the mathematical object being constructed — the Green's function of a coupled field manifold — is formally of the same *type* as the objects that arise in QFT, differing only in the dimensionality of the manifold and the nature of the probe. What was previously described as a structural analogy is here identified as a formal correspondence: a particle is a pole in the propagator of its field; a conscious emotional percept is a pole in the propagator of the soma-field. Different physics. Same mathematics.

That correspondence gives the model precise vocabulary for the following set of ideas, which are central to the clinical observation of emotion:

- A quantity that exists everywhere, continuously, even when unobserved
- A background of sub-threshold activity that is real and causally effective
- The emergence of observable phenomena (conscious feelings) through threshold-crossing excitation of that background
- The possibility of multiple simultaneous excitations that interact with one another

Note (*May 2026*)*:* A subsequent experiment (QUANT-EXP-1) demonstrates that the quantum extension of the Hopfield landscape used in this model — replacing the classical Langevin process with a transverse-field quantum annealer — produces a measurable *topological reachability advantage*: quantum annealing reaches attractor basins that cold classical dynamics cannot reach at any finite noise level. This upgrades the formal correspondence from a structural claim to a testable empirical prediction. See the companion paper *Quantum Soma and the Penrose Gap* (doi:10.5281/zenodo.20351230) for the full results and theoretical implications.

One further consequence follows. The clinical phenomena of alexithymia — difficulty identifying and naming feelings — and its apparent opposite, emotional flooding or hypervigilance, have always been treated as separate conditions requiring separate explanations. In the Green's function framing, they are the same structure at two extremes of the same parameter: the perception threshold T_i is too high (the bulk dynamics cannot cross into observable experience) or too low (bulk fluctuations flood the boundary without filtering). This is structurally identical to one of the deepest open problems in particle physics — the **hierarchy problem** — which asks why gravity is so much weaker than the other forces. The standard answer is that gravity propagates in the full higher-dimensional bulk while other forces are confined to a lower-dimensional brane; the coupling across the brane boundary determines the apparent weakness. The soma-field correspondence is exact: the threshold T_i *is* the brane. Perception is confined to the one-dimensional boundary of an eleven-dimensional dynamics. The hierarchy of emotional experience — why conscious feeling is so much weaker and more transient than the underlying field activity — has the same formal structure as the hierarchy of forces.

Neural Network Energy Functions and Hopfield Networks

In 1982, John Hopfield (awarded the Nobel Prize in Physics in 2024) proposed a model of associative memory based on a network of interconnected neurons (Hopfield, 1982). The critical insight was borrowed directly from statistical physics: the network could be assigned an **energy function** — a scalar quantity that decreases with each state update — such that the network would always evolve toward a local energy minimum. These minima are the stable states of the network: its memories, or more precisely, its *attractors*.

Hopfield observed that his neural network's dynamics were mathematically identical to those of an Ising spin-glass model from condensed matter physics — a system of interacting magnetic spins that minimises its total energy by aligning or anti-aligning with neighbours. The energy function he used is:

$$H(\mathbf{s}) = -\frac{1}{2}\sum_{i,j} W_{ij}\, s_i s_j - \sum_i \theta_i s_i$$

where $\mathbf{s}$ is the state of the network, W_{ij} is the coupling strength between units i and j, and θ_i is the activation threshold of unit i. The network always moves in the direction of decreasing H.

The Soma-Field Model applies this energy function directly to emotional dynamics. The *emotional coupling matrix* W encodes the relationships between emotional modes — which emotions amplify one another, which suppress one another — and the energy function determines the direction in which the emotional field naturally evolves.

Hopfield's network is a formal model of the *neocortex*: a system for storing cognitive patterns and retrieving them from partial cues by minimising an energy function. Every artificial neural network constructed since McCulloch and Pitts (1943) (McCulloch & Pitts, 1943) — from perceptrons to backpropagation networks to transformers — sits in this neocortical lineage. These systems recognise patterns, predict sequences, and minimise prediction error with increasing sophistication. None of them possess a limbic system. They have no internal valuation, no arousal modulation, no threat-detection architecture, no attachment structure, no interoception. They have very effective cortex.

The Soma-Field Model does not add to the neocortical lineage. It proposes the architectural layer that has never been formally built: *an artificial limbic system*.

Hopfield memory is associative and pattern-completing; somatic memory is state-reinstating. The field does not merely remember what happened. It re-lives it. *A body with a past.*

Hopfield's later-reported wish to have incorporated something analogous to 'maternal instincts' into the energy function was, in this reading, not a desire for a better cortex. It was an intuition pointing directly at the absent system — the layer beneath the cortex that assigns value, registers threat, and holds the body in a particular way of being long after the event that caused it.

This positions the Soma-Field Model not as a supplement to the neocortical lineage but as its completion. Artificial neural networks have, for eighty years, been increasingly sophisticated formal models of the neocortex: pattern recognition, sequence prediction, error minimisation. The cortex has been mapped in extraordinary detail. The limbic system — which assigns value, detects threat, modulates arousal, maintains attachment, and reinstates whole somatic states in response to partial cues — has had no comparable formal treatment. The architectural description of the vertebrate brain was, until this paper, half-built.

Four kinds of formal intelligence. This architectural gap can be situated within a wider taxonomy. Four quotients have been proposed to describe the landscape of biological intelligence across popular and scientific usage. They map onto the formal components of this model with an exactness that is not coincidental:

Quotient	What it measures	Biological substrate	Soma-Field status
IQ — cognitive	Pattern recognition, reasoning, prediction	Neocortex	Built (1943–): McCulloch & Pitts → Hopfield → transformers
EQ — emotional	Valuation, arousal, affect regulation	Limbic system	**Built here**: W, $K(\tau)$, $H(\mathbf{e})$, C_{HRV}, $\dot{H}$
AQ — adversity	Structural resilience under threat	PFC–limbic axis	**Built here**: S_{inst}, $\partial\|W\|/\partial t$, $C_{\text{HRV}}^{\text{recovery}}$
SQ — social	Attunement, theory of mind, relational navigation	Mirror system, TPJ	*Next paper*: κ_r, multi-field coupling

Table 3. Four dimensions of biological intelligence mapped onto the Soma-Field Model. The neocortical lineage (IQ) has been formally modelled for eighty years. Emotional intelligence (EQ) and adversity resilience (AQ) are formalised here for the first time. Social intelligence (SQ) is defined as the next extension of the framework.

AQ — adversity quotient — is formally the capacity to update W after adversity without the adversity permanently becoming W. Its mathematical definition appears in Section 3.4; its pathological lower bound is C-PTSD, in which all three components of AQ are simultaneously compromised (Appendix B.2).

The AI alignment implication follows directly. Current artificial systems have high IQ by construction and zero EQ, AQ, or SQ. The absence of internal valuation means that valuation must be injected externally — through reinforcement learning from human feedback (RLHF) and related techniques — which is structurally brittle for the same reason that a field with no limbic layer is brittle: the system has no internal stake in what it does. The Soma-Field formalisation specifies what that internal stake would look like, were it ever built.

A further lineage note is worth recording. Ramsauer et al. (2020) demonstrated that continuous-state modern Hopfield networks are mathematically equivalent to the self-attention mechanism in transformer language models. The softmax attention operation that drives contemporary large language models is a Hopfield retrieval step. The Soma-Field Model sits in this same energy-based lineage: the equations underlying associative memory, language understanding, and somatic trauma response are, at the appropriate level of abstraction, the same equations.

A historical irony completes the picture. String theory was not discovered as a theory of strings. In 1968, Gabriele Veneziano wrote down a scattering amplitude — a response function encoding how particles scatter — and only later did Nambu, Nielsen, and Susskind identify the string as whatever object produces that amplitude (Veneziano, 1968). The response function came before the thing. The Soma-Field Model recapitulates this historical order deliberately: the primary object is the eleven-dimensional coupling manifold; the string — the one-dimensional conscious percept — is what the manifold produces when probed. We retain Veneziano's discovery and decline to reify the string.

The Formal Correspondences: Where the Link Was Seen

The structural analogy between QFT and the Soma-Field Model is not merely conceptual. There are three places where equations from different disciplines become, after substituting the relevant quantities, literally the same functional form. The following sets them side by side. The point is not to impress with notation but to show exactly where the recognition happened — the moment when the same Greek letters appeared in the same positions in two fields that had no prior reason to be connected.

The same Hamiltonian: Ising spin model (condensed matter physics, 1920s) — Hopfield neural network (computational neuroscience, 1982) — Soma-Field Model:

$$H_{\text{Ising}}(\sigma) = -\frac{1}{2}\sum_{i,j} J_{ij}\,\sigma_i\,\sigma_j - \sum_i h_i\,\sigma_i$$

$$H_{\text{soma}}(\mathbf{e}) = -\frac{1}{2}\sum_{i,j} W_{ij}\,e_i\,e_j - \sum_i \theta_i\,e_i$$

Replace $J_{ij} \to W_{ij}$, $\sigma_i \to e_i$, $h_i \to \theta_i$: identical. The physicist, the neural network theorist, and the somatic clinician are computing the same energy function on different state spaces. The Hopfield 2024 Nobel Prize was awarded for discovering this identity between spin physics and neural computation; the Soma-Field Model extends the same identity one step further to emotional dynamics.

The Wick rotation — why the same exponential appears in QM and in memory:

In quantum mechanics, the time evolution operator is a complex phase:

$$U(t) = e^{-i\hat{H}t/\hbar}$$

Substitute $t \to -i\tau$ (the *Wick rotation* — replacing real time with imaginary time):

$$e^{-i\hat{H}(-i\tau)/\hbar} = e^{-\hat{H}\tau/\hbar}$$

The oscillating complex exponential becomes a real decaying exponential. This is the Boltzmann weight $e^{-\beta\hat{H}}$ at $\beta = \tau/\hbar$. The Langevin equation $\dot{\mathbf{e}} = -\nabla H + \eta$ is the classical limit of this Wick-rotated dynamics. Every simulation of the soma-field running this equation is, formally, a path integral in imaginary time.

The same propagator: Euclidean QFT (imaginary-time two-point correlator for a massive scalar field) — C-PTSD trauma memory kernel:

$$G_E(\tau) = \langle \phi(0)\, \phi(\tau) \rangle_{\text{QFT}} = \frac{1}{2m}\, e^{-m|\tau|}$$

$$K_{\text{trauma}}(\tau) = \sum_k A_k\, e^{-|\tau|/\tau_k}$$

Same form. The QFT field mass m corresponds to $1/\tau_k$ — the reciprocal of the trauma trace decay time. A heavier particle has a shorter-range propagator; a shorter-lived trauma trace decays faster. Therapeutic processing (reducing A_k, increasing τ_k) is, in the QFT language, changing the mass and amplitude of the propagator until the correlation function vanishes.

The specific visual moment: the quantum phase factor is $e^{-i\omega t}$. Remove the i (Wick rotation) and it becomes $e^{-\omega\tau}$. The memory kernel is $e^{-\tau/\tau_k}$. These are the same exponential. The i is the only difference between a quantum field that oscillates and a trauma trace that decays.

QFT quantity	Symbol	Soma-Field analogue	Symbol
Field mode	ϕ_k	Emotional mode	e_i
Coupling constant	J_{ij}	Coupling matrix entry	W_{ij}

QFT quantity	Symbol	Soma-Field analogue	Symbol
Field mass	m	Inverse decay time	$1/\tau_k$
Propagator amplitude	$1/2m$	Trauma trace amplitude	A_k
Euclidean propagator	$G_E(\tau) \propto e^{-m\tau}$	Memory kernel	$K(\tau) \propto e^{-\tau/\tau_k}$
Vacuum energy	$\langle H \rangle_0$	Resting field energy	$H(\mathbf{e}_{\text{calm}})$
Thermal fluctuation	$k_B T$	Noise amplitude	σ_0
Wick rotation	$t \to -i\tau$	Real-time Langevin	$\dot{\mathbf{e}} = -\nabla H + \eta$

Table 2. Formal correspondence between QFT quantities and Soma-Field analogues. Each row is a single mathematical entity in two notations. These correspondences were not constructed after the fact; they are the reason the QFT framework was recognised as relevant.

The central identification — particle and percept as poles in their respective propagators. All four correspondences above follow from one structural fact. In QFT, a particle is not a separate object from the field. It is a *pole* in the field's propagator — the Green's function evaluated in momentum space:

$$\tilde{G}_{\text{QFT}}(k^\mu) = \frac{i}{k^2 - m^2 + i\varepsilon}$$

The particle exists precisely when the four-momentum satisfies $k^2 = m^2$ — the *on-shell condition*. The particle is the singularity in the field's response to a point source: the field's Green's function, evaluated at its own resonance.

Diagonalise W with eigenvalues λ_i (the natural resonance frequencies of the emotional modes). The soma-field propagator — the two-point correlator $\langle e_i(t)\, e_i(t')\rangle$ in the frequency domain — is:

$$\tilde{G}_{ii}(\omega) = \frac{\sigma_{\text{eff}}^2}{\omega^2 + \lambda_i^2}$$

A conscious emotional percept in mode i exists precisely when the excitation frequency ω approaches $i\lambda_i$ — the mode's natural resonance. The percept is the singularity in the soma-field's response to a somatic probe.

Setting the two propagators side by side:

$$\frac{i}{\underbrace{k^2 - m^2 + i\varepsilon}_{\text{QFT: particle at mass-shell } k^2=m^2}} \quad \longleftrightarrow \quad \frac{\sigma_{\text{eff}}^2}{\underbrace{\omega^2 + \lambda_i^2}_{\text{Soma-Field: percept at resonance } \omega=i\lambda_i}}$$

Both are poles in the propagator of their respective field manifold. A photon is not the electromagnetic field; it is the field's Green's function evaluated at a resonance. A flash of conscious emotion is not the soma-field; it is the field's Green's function evaluated at a threshold-crossing resonance. The manifolds differ — one is the four-dimensional spacetime vacuum, the other is the eleven-dimensional emotional coupling geometry. The mathematical type is the same. This is not analogy.

The Body Schema, Interoception, and Pain

A complete model of the emotional field must address a phenomenon that standard psychological accounts of emotion consistently underspecify: the field is not a model of the physical body. It is the nervous system's *predictive model* of the body — a continuously updated internal representation of what the soma should be experiencing, revised by incoming interoceptive signals.

The clinical proof of this distinction is phantom limb pain (Ramachandran & Hirstein, 1998). Patients who have undergone amputation routinely experience pain in the absent limb. The pain is real: it activates the same neural circuits, produces the same suffering, and responds to the same analgesics as pain from an intact limb. The limb is gone. The neural model of the limb persists. What hurts is the *brain's representation* of the foot, not the foot.

This is not an anomaly. It is the normal condition of all somatic experience. The brain does not receive raw signals from the body — it maintains a continuous predictive model of the body (the *body schema*) and generates somatic experience from that model. Interoception — the sense of the internal body state — is a prediction, not a direct readout (Seth, 2021). The brain predicts what the heart should be doing, what the gut should feel like, where tension should be. The felt body is the predicted body.

The formal consequence is direct: the soma-field's state vector $\mathbf{e}(t)$ must include **somatic modes** — pain states, regional tension, visceral sensation, proprioceptive activation — alongside emotional modes. These are modes of the same field, governed by the same coupling matrix W. The W_{ij} between fear modes and somatic pain modes is the formal account of why fear amplifies pain, why safety reduces it, and why chronic pain and C-PTSD are highly comorbid. They are not separate conditions sharing a correlation. They are the same attractor architecture operating across emotional and somatic modes simultaneously.

Phantom limb as attractor persistence. An amputated limb's somatic modes do not disappear from W when the limb is removed. The neural model persists. When movement- intention modes are activated — attempting to move the absent foot — foot-sensation modes are co-activated via W. If co-activation exceeds threshold, it is experienced as pain. Ramachandran's mirror box provides visual input that disconfirms the prediction error: new sensory evidence that the limb is moving, reducing coupling-driven co-activation, and therefore reducing the pain. This is $W \to W'$: therapy as structural rewriting of the field.

The load-bearing hyphen. The term *emotional-somatic* in clinical literature is not a stylistic compound. The hyphen marks an ontological claim: emotional states and somatic states are not two separate things that correlate. They are two aspects of the same field. The coupling matrix W is precisely the hyphen, made formal.

Therapeutic implication. Somatic therapies — body scanning, sensorimotor work, EMDR's bilateral stimulation — work not on the physical body but on the brain's model of the body. They provide new interoceptive evidence that updates the prediction. They change W. Therapy does not fix the tissue. It updates the model.

Correspondence with Existing Emotion Representations

A reasonable objection to any new framework is: *there is already a great deal of structure out here.* This is true. The emotion research literature contains several well-developed representational systems, and the Soma-Field Model must be positioned relative to them. The short answer is that every existing representation is *descriptive*; the Soma-Field Model is *dynamical*. The longer answer follows.

Categorical taxonomies (Ekman 1972; Plutchik 1980; Parrot 2001) assign names and hierarchical membership to emotional states. They are ontologies in the formal sense: a T-Box of classes and subclass relations. Plutchik's wheel additionally defines a *blend* operation — Love := Joy ⊓ Trust, Awe := Fear ⊓ Surprise — which is precisely the OWL2 `intersectionOf` construction. These systems tell you what to call a state. They do not tell you how a state evolves, or which attractor a system settles in when two mechanisms fire simultaneously.

Dimensional models (Russell 1980; Mehrabian and Russell 1974) embed emotions in a continuous space, canonically Valence × Arousal (the *circumplex*), sometimes extended to Pleasure × Arousal × Dominance. These models capture the *coordinates* of a state. The energy landscape of the Soma-Field Model — the function $H(\mathbf{e})$ over emotion-space — is the dynamical generalisation of the circumplex: the circumplex is a snapshot of positions; the energy landscape is the surface over which the field moves. The stable attractors of H are the emotion categories; their coordinates are the circumplex positions.

Process and appraisal models (Scherer 1999; Frijda 1986; the OCC model of Ortony, Clove and Collins 1988) describe the *sequence of evaluations* through which a stimulus becomes an emotion. They are closer to the Soma-Field dynamics — they include temporal stages — but they are deterministic and single-threaded: one appraisal chain, one output. The Soma-Field replaces this with a parallel field update: all modes evolve simultaneously, governed by the full W matrix.

Music-specific schemas (BRECVEMA, Juslin and Västfjäll 2008; Juslin *et al.* 2011; GEMS, Zentner *et al.* 2008) are the closest antecedents to the present model. The BRECVEMA framework identifies eight distinct psychological mechanisms through which music evokes emotion — Brain stem reflex, Rhythmic entrainment, Evaluative conditioning, Contagion, Visual imagery, Episodic memory, Musical expectancy, Aesthetic judgement — each with distinct evolutionary origins, processing speeds, and neural substrates. These mechanisms are the *object properties* of the emotion-induction ontology: they specify which musical features activate which emotional outputs. Juslin explicitly identifies the open problem: *"Exploring how various musical emotions come about through the interaction of multiple psychological mechanisms is an exciting endeavour that has just begun"* (Juslin & Sloboda, 2011, p. 638). The W coupling matrix is the formal answer to that open problem. Where BRECVEMA gives a list of mechanisms with characteristic outputs, the Soma-Field gives the interaction tensor W_{ij} that specifies, with numerical precision, what happens when mechanisms i and j fire concurrently.

The deeper connection is spectral. The *eigenmodes* of W — the directions in emotion-space that evolve independently — are the natural resonances of the soma-field: the patterns the field rings with when struck. BRECVEMA mechanisms are inputs: they excite specific rows of W. The eigenspectrum of W is the response: the set of frequencies the manifold can sustain. Where BRECVEMA is a taxonomy of *stimuli*, the eigenspectrum of W is a taxonomy of *responses*. Juslin's open problem — how mechanisms interact — is the question of how stimulus-space maps onto eigenmode-space through W. Section 3.3 develops this.

Body maps (Nummenmaa *et al.* 2014) map emotions to their somatic distribution — where in the body each emotion is felt. These are precisely the spatial support of the soma-field modes: the field configuration corresponding to an attractor state is the body map of that emotion. Body maps are measurements of the attractors; the Soma-Field is the dynamical system that generates them.

The formal correspondence table extends Table 2 to include these systems:

Existing representation	What it captures	Soma-Field equivalent
Ekman categories	Attractor labels (names)	Values of $\mathbf{e}$ at energy minima
Plutchik dyads ($A \sqcap B$)	Blend attractors	Metastable states between two energy minima
Russell circumplex	Coordinates (valence, arousal)	Projection of $H(\mathbf{e})$ onto two axes
OCC appraisal tree	Single-path sequential process	Single trajectory in the full field
BRECVEMA mechanisms	Object properties: stimulus → emotion	Rows of W: mechanism i activates mode j
Body maps (Nummenmaa)	Spatial support of each attractor	Modal structure of $\mathbf{e}$ at each minimum

None of these correspondences require modifying either the existing representations or the Soma-Field Model. They are consequences of the model's structure. The formal machinery for exploring these correspondences — typing BRECVEMA mechanisms as Lean inductive constructors, Plutchik blends as type intersections, mechanism profiles as decidable propositions — is developed in the companion file `src/EmotionOntology.lean`.

The Soma-Field Model

The field is primary. The felt emotion is secondary — it is what registers when the field is probed. This is the same ontological relationship as between a quantum field and a particle: the field exists continuously and everywhere; the particle is what you observe at the moment of measurement. The Soma-Field Model does not describe what emotions are *made of*. It describes the manifold whose impulse response *is* conscious emotional experience.

Emotions as a Persistent Wave Field

The foundational claim of the Soma-Field Model is simple: emotions are not events. They are a *field* — a distributed, continuous quantity defined over the entire soma (body-mind system) at all times.

This field has two coupled components:

1. **The somatic wave** $\mathbf{E}_{\text{body}}(x,t)$: distributed across the body as patterns of visceral sensation, muscle tone, proprioception, interoception, and autonomic state.
2. **The neural wave** $\mathbf{E}_{\text{neural}}(x,t)$: distributed across the nervous system as patterns of activation in cortical, subcortical, and peripheral neural circuits.

These two components are not separate systems. They are coupled — each continuously influencing the other. The total emotional field is their combined state:

$$\mathbf{E}(x,t) = \mathbf{E}_{\text{body}}(x,t) \otimes \mathbf{E}_{\text{neural}}(x,t)$$

The field is characterised by:

- **Multiplicity**: multiple emotional modes can be simultaneously active and interfering
- **Continuity**: it exists at all times, not only during episodes of conscious feeling
- **Spatial distribution**: different aspects of the field are localised in different regions of the soma (the familiar clinical observation that grief is felt in the chest, fear in the gut, anger in the jaw and fists)
- **Temporal dynamics**: the field evolves continuously, driven by the energy function

Figure 1. The Soma-Field. The body and brain are not separate containers of emotion but two coupled components of a single distributed wave field. Neither is primary; each continuously modifies the other. The ≈ symbols indicate that wave activity is always present in each region, not only during episodes of conscious feeling.

The Perception Threshold

Not all activity in the emotional field is consciously perceived. The field has a **perception threshold** T_i for each emotional mode i. Below this threshold, the emotional mode is sub-perceptual: it exists, it influences behaviour and physiology, but it does not surface as a named conscious feeling.

$$\text{Emotion } i \text{ is consciously perceived} \iff |\mathbf{E}_i(t)| > T_i$$

This threshold crossing corresponds precisely to the QFT excitation analogy: the emotional mode behaves like a virtual particle that has accumulated enough energy to become real — to emerge from the sub-threshold background and enter awareness.

This accounts for a range of clinically significant phenomena:

Clinical Observation	Soma-Field Account
Patient reports no feeling but shows physiological signs of distress	Sub-threshold field activity below T_i
Sudden unexpected flood of emotion in session	Rapid threshold crossing after gradual accumulation
Emotion felt somatically but not named	Threshold crossed in $\mathbf{E}_{\text{body}}$, not yet in $\mathbf{E}_{\text{neural}}$
Alexithymia (difficulty identifying feelings)	Elevated T_i — high threshold requiring more energy to cross
Hypervigilance / emotional flooding	Lowered T_i — reduced threshold, field crosses to conscious easily

Table 1. Clinical observations mapped onto the perception threshold model.

Figure 2. The perception threshold T_i for a single emotional mode. The field is active continuously (lower trace). Conscious experience arises only when amplitude exceeds T_i (upper trace). Everything below the line is still there — shaping body and behaviour before it can be named.

Figure 0. Continuous soma-field activity (blue) with a single threshold-crossing event. The field is always active; conscious experience (shaded) arises only when amplitude exceeds the perception threshold θ (red dashed). Below the threshold: real, causally active, but not yet conscious.

The Interaction of Emotional Modes

Multiple emotional modes are simultaneously active in the field at all times. They do not simply co-exist: they interact. The nature of these interactions is encoded in the **emotional coupling matrix** W, where W_{ij} represents the influence of emotional mode j on emotional mode i.

- If $W_{ij} > 0$: emotion j amplifies emotion i (e.g., fear can amplify shame)
- If $W_{ij} < 0$: emotion j suppresses emotion i (e.g., calm suppresses anxiety)
- If $W_{ij} = 0$: emotions i and j are independent

The field evolves according to the energy gradient:

$$\dot{\mathbf{e}}(t) = -\nabla H(\mathbf{e}) + \eta(t)$$

where $\eta(t)$ represents the continuous low-level fluctuations of the sub-perceptual field — the emotional equivalent of quantum vacuum noise. The field is always moving, always seeking lower energy, never at absolute rest.

The Three-Layer Architecture

The nervous system that implements the soma-field is not architecturally flat. Three hierarchically organised layers contribute to field dynamics, each corresponding to a distinct evolutionary substrate and a distinct role in the model. The clinical literature (Porges, 2011; van der Kolk, 2014; Ogden et al., 2006) converges on this stratification; what follows is its formal expression.

Layer 1 — Brainstem / autonomic baseline. The oldest structures: vagal nuclei, arousal systems, interoceptive machinery. In the model, this layer is represented by the noise term and, specifically, by the heart rate variability coherence C_{HRV}, which modulates effective noise amplitude across the whole field:

$$\sigma_{\text{eff}} = \frac{\sigma_0}{C_{\text{HRV}}}$$

High HRV coherence narrows effective noise, stabilising the field in its current attractor. This is the mechanism of HRV biofeedback as a regulatory intervention: it does not target any specific emotional mode but lowers the fluctuation floor of the entire field.

Layer 1 extension: cardiac acceleration and landscape tilt. The term C_{HRV} measures the *current state* of cardiac regularity — where the heart is. A complementary quantity is $\dot{H}(t)$, the first time-derivative of heart rate, in units of beats/s^2. This is the **cardiac acceleration**: not what the heart rate is, but where it is going.

The dimensional parallel with gravity is exact: gravitational acceleration g carries units m/s^2; cardiac acceleration $\dot{H}$ carries units beats/s^2. Both are accelerations; both describe a force field rather than a position. Gravity does not tell you where a test mass is — it tells you how it will move next. Cardiac acceleration tells you not the current BPM but the direction of the next one: the N+1 state.

In the soma-field, $\dot{H}(t)$ enters the dynamics not as noise modulation but as a **landscape tilt** — a time-varying bias added to the Hamiltonian that tips the energy function toward activation or rest attractors:

$$H(\mathbf{e}, t) = H_0(\mathbf{e}) - \alpha\,\dot{H}(t)\,\beta \cdot \mathbf{e}$$

where $\alpha > 0$ is the cardiac-somatic coupling constant and β is a mode-coupling vector (at leading order, $\beta = \mathbf{1}$: the tilt acts uniformly across all modes). When $\dot{H}(t) > 0$ (heart accelerating), the landscape tilts toward higher activation states before any cognitive or affective threshold is crossed. When $\dot{H}(t) < 0$ (heart decelerating), it tilts toward rest. The full three-layer equation including the cardiac acceleration term is:

$$\dot{\mathbf{e}}(t) = -\nabla H_0(\mathbf{e}) + \alpha\,\dot{H}(t)\,\beta + R_{\text{PFC}}(\mathbf{e}, t) + \frac{\sigma_0}{C_{\text{HRV}}}\,\xi(t)$$

The two cardiac terms serve distinct functions: C_{HRV} (state) modulates the noise floor; $\dot{H}$ (acceleration) tilts the deterministic landscape. Both are needed for a complete account of cardiac influence on the field.

Predictive clinical value. A patient with BPM = 90 and $\dot{H} = +4$ beats/s^2 is approaching threshold; one with BPM = 90 and $\dot{H} = -4$ beats/s^2 is retreating from it. The snapshot is identical; the trajectories are opposite. Cardiac acceleration is therefore an early-warning signal for threshold crossings — detectable at Layer 1 before the emotional field at Layer 2 has crossed its threshold. This has independent support in cardiology: Bauer et al. (2006) demonstrated that *acceleration capacity* and *deceleration capacity* of heart rate — estimates of $\dot{H}$ over a cardiac window — carry prognostic information independent of conventional HRV measures.

The somatic equivalence principle. The cardiac acceleration term $\alpha\,\dot{H}\,\beta$ is structurally identical in the equation to any other forcing term. From the perspective of the field itself — from conscious experience — cardiac-driven activation is indistinguishable from event-driven activation. A sudden heart rate acceleration tilts the landscape by exactly the same mechanism as an external threat or an intrusive memory. The field has no access to the origin of the tilt. This is the formal account of a clinically well-documented phenomenon: anxiety initiated by cardiac irregularity (arrhythmia, postural hypotension, caffeine, exertion) is experienced as emotionally caused, because the somatic signal is identical. Disambiguation requires either external measurement or deliberate interoceptive inquiry that can distinguish the two sources.

Layer 2 — Limbic system / emotional memory. The primary substrate of the Soma-Field Model. The coupling matrix W, memory kernel $K(\tau)$, Hamiltonian $H(\mathbf{e})$, and threshold T all belong here. The limbic layer stores emotional-somatic states and reinstates them in response to partial body cues: a continuous, asymmetric, temporally extended Hopfield network operating on somatic states rather than cognitive patterns. This is the architectural layer that has been absent from every artificial neural network since McCulloch and Pitts (1943) (McCulloch & Pitts, 1943). The cortex has been modelled many times; the limbic system has not.

Structural plasticity under adversity. The Soma-Field framework permits a formal characterisation of the field's resilience under adverse conditions. Define the *plasticity index* Π as a composite of three measurable field properties:

$$\Pi \;=\; \frac{1}{S_{\text{inst}}} + \left.\frac{\partial \|W\|}{\partial t}\right|_{\text{adversity}} + C_{\text{HRV}}^{\text{recovery}}$$

The three terms correspond to: (i) how accessible regulated-state attractors remain under adversity ($1/S_{\text{inst}}$, instanton accessibility — Section 4.4); (ii) how much the coupling matrix can structurally adapt following a threshold crossing ($\partial\|W\|/\partial t$, the plasticity component); and (iii) how quickly the HRV floor recovers after activation ($C_{\text{HRV}}^{\text{recovery}}$, the regulatory resilience component). Complex PTSD is the clinical presentation of chronically low Π across all three terms simultaneously: high barriers to regulated attractors, a rigid W dominated by threat configurations, and impaired C_{HRV} recovery. Structural plasticity is the capacity of the field to update W in the aftermath of adversity without the adversity permanently *becoming* W.

Layer 3 — Neocortex / prefrontal regulatory layer. Top-down modulation of Layer 2, represented as a regulatory term $R_{\text{PFC}}(\mathbf{e}, t)$. The full field dynamics becomes:

$$\dot{\mathbf{e}}(t) = -\nabla H(\mathbf{e}(t)) + R_{\text{PFC}}(\mathbf{e}, t) + \frac{\sigma_0}{C_{\text{HRV}}}\,\xi(t)$$

R_{PFC} represents voluntary attention, therapeutic technique, and conscious reappraisal acting on the field. It is not a correction of Layer 2 but a modulation of it. Under sustained therapeutic engagement, R_{PFC} participates in the structural modification $W \to W'$ constituting the forward transformation (Section 7).

The **threshold T is the Layer 2 / Layer 3 boundary**: sub-threshold dynamics are processed limbically and remain below conscious awareness; threshold-crossing events enter Layer 3 and become available for narrative, meaning-making, and voluntary response. This is the formal basis for the clinical observation that insight without somatic activation is limited, and somatic activation without Layer 3 engagement cannot produce structural change: the layers are coupled, not independent. R_{PFC} requires a threshold crossing in order to have something to work with.

The two-term Langevin equation introduced in Section 3.3 is the Layer 2 special case ($R_{\text{PFC}} = 0$, $C_{\text{HRV}} = 1$). All subsequent sections develop that special case. The full three-layer equation is the general form.

The Energy Landscape

The Structure of the Emotional Energy Function

The energy function $H(\mathbf{e})$ defines a landscape over the space of possible emotional states. Like a physical landscape of hills and valleys, this landscape has:

- **Valleys (local minima)**: stable emotional states the field naturally moves toward
- **Hills (local maxima)**: unstable configurations the field naturally moves away from
- **Saddle points**: transitional configurations with mixed stability

The key property of an energy function is directionality: the field always moves *downhill*. It always evolves toward lower energy. Therapeutic intervention, in this framework, can be understood as:

1. **Changing the landscape**: modifying W — the coupling matrix — through new relational experience, insight, or somatic work, so that the energy minima are in healthier locations
2. **Adding energy to escape a trap**: helping the field accumulate enough energy to escape a deep but unhealthy local minimum (e.g., the freeze state)
3. **Pointing toward the global minimum**: orienting the field toward regulated calm

Attractor States: Fight, Flight, Freeze, and Regulated Calm

The Soma-Field Model proposes that the major attractor basins of the emotional energy landscape correspond directly to the autonomic states described by Porges' polyvagal theory.

Figure 3a. Topographic (bird's-eye) view of the energy landscape. The field always rolls downhill toward the nearest minimum. Freeze and calm are both low-energy — but freeze is surrounded by high walls. Escape from freeze requires crossing those walls, which means first gaining energy before losing it again. This is the clinical challenge of working with dissociative states.

```
ENERGY
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  |        |  |  |  |
  |   _____|  |  |  |_____
  |  |        \/        |
  |  |      saddle       |
  |  |    (transition)   |
```

```
  │  │                │    ╔═════════╗
  │  │      freeze    │    ║         ║
  │  │       ┌─┐      │    ║regulated║◄── global minimum
  │  │_______│ │______│    ║  calm   ║
  │          │             ╚═════════╝
  └────────────────────────► EMOTIONAL STATE SPACE
```

Figure 3b. Schematic energy landscape. Fight/flight are high-energy, unstable local minima. Freeze is a low-energy but isolated attractor — easy to enter, hard to escape. Regulated calm is the global energy minimum.

Attractor	Energy State	Polyvagal Correlate	Clinical Presentation
Regulated Calm	Global minimum	Ventral vagal (social engagement)	Present, flexible, connected
Fight	Shallow high-energy minimum	Sympathetic (mobilisation)	Agitation, anger, urgency
Flight	Saddle point / shallow minimum	Sympathetic (mobilisation)	Anxiety, avoidance, rumination
Freeze	Deep isolated minimum	Dorsal vagal (immobilisation)	Dissociation, numbness, collapse

Table 2. Emotional attractors mapped onto Polyvagal states.

The therapeutic significance of this structure is considerable. The freeze state is dangerous not because it is high-energy — it is in fact very low energy — but because it is *isolated*: surrounded by energy barriers that make it difficult to exit. Escape from freeze requires first *increasing* the field's energy (mobilising some arousal) before it can flow toward regulated calm. This corresponds well to the clinical observation that working with dissociated patients requires careful titration of arousal — not too much, not too little — before emotional processing is possible.

The Coupling Matrix as a Personal Signature

The coupling matrix W is not universal. Each person has a unique W, shaped by attachment history, trauma, cultural context, and temperament. A person with a history of developmental trauma may have a W in which anxiety and shame are strongly coupled ($W_{\text{shame, anxiety}} \gg 0$), creating a combined attractor that is particularly deep and sticky. A person with a secure attachment history may have a W in which positive emotions are broadly coupled to one another, creating a wide basin around regulated calm.

This implies that the energy landscape is a therapeutic object in its own right: understanding a patient's W is understanding the structural dynamics of their emotional life.

In the M-theory compactification analogy developed in Appendix A, the coupling topology W corresponds to the shape of the compact G_2 manifold — the seven-dimensional geometry that determines which force-like couplings are allowed and with what strengths. That analogy is here made precise: two people differ not merely in their emotional *parameter settings* but in their coupling *geometry*. Developmental trauma does not set a dial to the wrong value; it deforms the manifold. The therapeutic process of modifying W through relational experience, insight, or somatic work is, in this language, differential geometry: a continuous deformation of the G_2 manifold toward a configuration in which the regulated-calm attractor is globally accessible. The practitioner is, without having been told so, a geometer.

Dissonance and Resolution

The Acoustic Analogy

The Soma-Field Model draws a further structural analogy, this time with acoustics. When two sound waves interact, the quality of their interaction — consonance or dissonance — depends on the phase relationship between them. Consonant intervals (the octave, the fifth) have simple frequency ratios and produce stable, reinforcing interference patterns. Dissonant intervals (the tritone, the minor second) have complex ratios and produce beating, unstable, tension-generating patterns.

The model proposes that the same relationship holds between emotional modes. When two emotional modes are in a compatible relationship — when their interaction is consonant — the field is in a relatively low-energy configuration and moves naturally toward the energy minimum. When they are in an incompatible relationship — when their interaction is dissonant — the field is in a higher-energy configuration, generating a gradient that drives toward resolution.

Dissonance, in this framework, is felt as tension. It is not pathological; it is directional. Dissonance is the field's way of communicating that it is far from equilibrium and that resolution is available.

The Resolution Principle

In music, dissonance resolves to consonance. The tritone — the most dissonant interval in Western tonality — creates a powerful gravitational pull toward resolution. In counterpoint, the rules of voice leading describe the specific paths by which dissonance must resolve. These rules are not arbitrary conventions; they describe the geometry of the acoustic energy landscape.

The same principle applies to emotional dissonance. An unresolved emotional state — grief that has not been fully experienced, anger that has been suppressed, fear that has been dissociated — is a dissonance in the field. It generates a persistent tension gradient. The therapeutic process can be understood as guided voice leading: finding the specific path of resolution that transforms the dissonant configuration into a consonant one.

This provides a formal basis for a widely-held clinical intuition: that emotions need to be *felt through* rather than avoided. Avoidance keeps the field in a dissonant state. The energy minimum — regulated calm — lies on the other side of the dissonance, not around it.

The Soma-Field Instrument

Rationale

The Soma-Field Model is not only a theoretical framework. It motivates a practical therapeutic instrument: a means by which a person can *externalise* their emotional field — make it visible and audible — and interact with it in real time.

The core insight is that the emotional field is normally invisible to its host. It operates below the threshold of conscious awareness, shaping behaviour and physiology without being available for reflection. If its activity could be rendered as a signal — a sound, an image, a pattern — it could become an object of therapeutic attention.

Design

The instrument uses a MIDI controller with 16 rotary knobs as its input interface. Eight emotional dimensions are encoded, each represented by two knobs:

- **Knob 1** of each pair: the somatic (body-level) intensity of that emotional mode
- **Knob 2** of each pair: the cognitive/neural intensity of that emotional mode

This design reflects the two-component structure of the field: body and mind are encoded separately but coupled in the computation. Each knob has a continuous range, allowing fine expression of emotional intensity.

```
┌──────────────────────────────────────────┐
│         MIDI CONTROLLER                  │
│                                          │
│  [K1][K2]  [K3][K4]  [K5][K6]  [K7][K8]  │
│  emotion1  emotion2  emotion3  emotion4│
│                                          │
│  [K9][K10] [K11][K12][K13][K14][K15][K16] │
│  emotion5  emotion6  emotion7  emotion8│
└──────────────────────────────────────────┘
                    │
                    ▼
           ┌────────────────────┐
           │  ENERGY FUNCTION   │
```

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│  H(e) computed    │
│  □H(e) computed   │
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          │
┌─────────┼─────────┐
▼         ▼         ▼
AUDIO OUTPUT        MIDI OUTPUT         VISUAL OUTPUT
(timbre reflects    (pitch/velocity     (field map:
dissonance)          reflects energy)    wave topology)
```

Figure 4. The Soma-Field Instrument: input, computation, and multimodal output.

The Feedback Loop

The instrument creates a **closed feedback loop** between the person and their emotional field:

1. The person expresses their current emotional state by adjusting the knobs
2. The system computes the energy function $H(\mathbf{e})$ and its gradient ∇H
3. The energy level, dissonance, and proximity to attractor states are rendered as:
   - **Sound**: harmonic content and timbre reflect the consonance or dissonance of the current state
   - **MIDI output**: pitch rises with tension, resolves as energy decreases
   - **Visual**: a real-time map of the emotional field, showing wave activity, threshold crossings, and the direction of the energy gradient
4. The person hears and sees their emotional field, and adjusts the knobs in response

This loop externalises the emotional field's gradient — the direction in which it is *trying* to move — and makes it available as sensory information. The person becomes not only the source of the emotional signal but also its observer, creating the conditions for reflection and regulation that are at the heart of therapeutic work.

The Pluggable Emotion Model

No single model of the emotions is assumed. The coupling matrix W — the structure that determines how emotional modes interact — is loaded from an external configuration file. Standard models (Plutchik's wheel of emotions, Ekman's basic emotions, the valence-arousal-dominance dimensional model) are provided as defaults. The therapist or client can modify the coupling values to reflect their own understanding of their emotional patterns, or a new model can be substituted entirely. The computational engine is model-agnostic.

Clinical Implications

Assessment

The Soma-Field Model suggests a different orientation for emotional assessment. Rather than asking "What emotion do you feel?" — which presupposes threshold-level conscious awareness — it invites attention to the sub-perceptual field: "What is present in the body right now, even if you cannot name it?" This aligns with Focussing-oriented approaches and with sensorimotor methods that prioritise somatic signal over narrative content.

The energy landscape provides a clinical map. A person chronically in a fight or flight attractor shows a different energy signature from a person in a freeze attractor, even if their presenting narratives are superficially similar. The model suggests that these are structurally different therapeutic challenges: fight/flight require down-regulation, while freeze may first require careful up-regulation before down-regulation becomes possible.

Intervention

The energy function provides a formal basis for several existing clinical interventions:

- **Grounding and titration** (Levine, 2010): adding small, controlled amounts of energy to the field to approach — without flooding — a previously frozen or avoided emotional state
- **Pendulation** (Levine, 2010): oscillating between a dissonant state and a resource state, progressively widening the tolerance window — equivalent to approaching the energy minimum via a series of small excursions
- **Somatic resourcing** (Ogden et al., 2006): establishing a stable low-energy region in the landscape that the field can return to after excursions into high-energy territory
- **Working with the felt sense** (Gendlin, 1978): attending to sub-threshold field activity and allowing it to cross the perception threshold in a supported context

Psychoeducation

The wave model is immediately accessible to clients who have struggled to understand their emotional experience. The statement: *"Your emotions are like waves — they are always there, even when you can't feel them, and they are always moving"* is both technically accurate within the Soma-Field framework and clinically useful as a normalising frame for sub-threshold emotional activity, for the apparently sudden onset of strong feelings, and for the experience of feeling multiple conflicting emotions simultaneously.

The energy landscape metaphor — *"right now the field is in a valley that is hard to leave, but it is not the lowest valley available to you"* — offers a way to discuss the freeze state, dissociation, and emotional stuckness without pathologising, while still acknowledging the structural difficulty of these states and the work required to shift them.

Neurodivergent Conditions as Operator Modifications

A clinically significant extension of the Soma-Field Model concerns neurodivergent conditions — specifically Autism Spectrum Condition (ASC), Attention Deficit Hyperactivity Disorder (ADHD), and Complex Post-Traumatic Stress Disorder (C-PTSD), which frequently co-occur and each present distinct challenges for somatic emotional processing.

The key architectural principle is this: **these conditions are not parameter settings in the model. They are structural modifications to the operators themselves.** This distinction matters both mathematically and clinically. A parameter change ("set the fear threshold lower") is a quantitative adjustment within the existing structure. An operator modification changes the *form* of the dynamics — it alters the governing equations, not merely their coefficients. Each condition wraps the standard pipeline in a different functional modifier, and — critically for the many individuals who carry all three — these modifiers *compose*. The combined condition is not three separate problems; it is the composition of three operators acting on the same underlying field.

The mathematical details of each modifier are given in Appendix B. Clinically, the consequences are as follows.

Complex PTSD introduces a *memory kernel* into the field dynamics: past high-energy states leave decaying echoes that continue to excite the field without new external stimulus. This is why traumatic activation can appear without identifiable trigger — the field is responding to its own history, not its current environment. The standard Hopfield attractor topology is also disrupted: C-PTSD renders the freeze attractor pathologically deep and wide, the window of tolerance (the basin around regulated calm) pathologically narrow, and the coupling matrix asymmetric — a condition under which the field can enter persistent *limit cycles* rather than settling to a stable minimum. Re-experiencing, flashbacks, and hypervigilance are, in this framework, limit-cycle oscillations in the traumatised field.

```
REGULATED FIELD  (symmetric W, no memory kernel)
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```

```
    ↑ baseline returns to near-zero between episodes
    ↑ each threshold crossing is a discrete, independent event
    ↑ 'regulated calm' is a genuine resting state — the global energy minimum


  C-PTSD MODIFIED FIELD  (asymmetric W, memory kernel K(t-s) present)
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  ||           \/    \/            \/    \/
  ||  ← even the troughs stay near T or above: baseline is elevated
  └────────────────────────────────────────────────────────────────────────► t
    ↑ memory kernel: each activation feeds energy back into the next
    ↑ field rarely returns to true rest — past states re-enter present dynamics
    ↑ almost entirely above T: activation is the default, not the exception
    ↑ 'regulated calm' requires a non-perturbative transition (the instanton):
      small steps do not reach it; a qualitatively different move is needed
```

Figure 5. The same emotional field mode under two dynamic regimes. Top: regulated dynamics — the field oscillates and returns to a low baseline between episodes; conscious emotion (above T) is episodic and resolves. Bottom: C-PTSD-modified dynamics — the memory kernel elevates the baseline so that the field rarely returns to rest; episodes bleed into one another; the system cycles rather than settles. The mathematical basis for this comparison is given in Appendix B.2.

Developmental timing and what can be recovered. The character of the C-PTSD modification depends critically on *when* it occurred — the developmental age τ_d at which the primary traumatic modification took place.

For **late trauma** (τ_d large — adult or post-verbal): a coupling matrix W_0 formed before the event. The modification is additive: $W = W_0 + \delta W_{\text{trauma}}$. A counterfactual pre-trauma self exists, encoded in explicit narrative memory. Therapeutic processing can target δW specifically, and the goal of recovering proximity to W_0 is formally coherent.

For **early trauma** (τ_d small — pre-verbal, perinatal): the coupling matrix W was *formed under the modification*. There is no W_0. The asymmetric coupling and the memory kernel coefficients are the baseline architecture, not additions to one. A counterfactual pre-trauma self was never encoded — it does not exist as a recoverable state.

This is a formal statement of a clinical fact that somatic therapists recognise but rarely have a mechanistic basis for: early trauma cannot be *processed away* in the sense of recovering a prior self, because no prior self was formed. The therapeutic goal is not subtraction ($W \to W_0$, which is undefined) but **forward transformation**: constructing a W' that supports a wider window of tolerance, different attractor topology, and lower memory-kernel amplitudes. This is a different mathematical operation — and requires a different therapeutic model.

The Soma-Field Instrument can reflect this distinction directly: a user whose primary modification is pre-verbal initialises with a *structural* coupling matrix (the modification *is* the baseline), not a neurotypical matrix with an added modifier. The formal basis for this parameterisation is given in Appendix B.2.1.

ADHD raises the effective *thermal noise* of the field — the amplitude of the sub-perceptual fluctuations — and simultaneously reduces the damping coefficient that slows the field's response to the energy gradient. The result is a field that explores its energy landscape rapidly and unpredictably, is easily displaced from shallow attractor basins by small perturbations (distractibility), but also achieves states of intense concentration (hyperfocus) when the coupling to a high-salience stimulus temporarily deepens a specific attractor basin far beyond its resting depth. ADHD is not a deficit of attention; it is a high-temperature, low-damping emotional field with a stimulus-dependent attractor structure.

Autism Spectrum Condition modifies the *projection kernels* — the functions that determine how the continuous somatic field is sampled to produce the discrete state vector — and the *sparsity* of the coupling matrix. Interoceptive research in autism (Garfinkel et al., 2016) documents significant differences in the processing of internal body signals; in model terms, certain somatic regions are over-represented (heightened sensory sensitivity) and others under-represented (reduced interoceptive clarity, contributing to alexithymia). The coupling matrix in ASC tends toward greater sparsity — fewer strong cross-modal emotional couplings — a pattern consistent with monotropism (Murray, 2018): the field settles deeply into individual attractors but transitions between them require proportionally more energy. Intense interests, emotional consistency within a context, and difficulty with unexpected transitions all follow from this attractor topology.

For the Soma-Field Instrument, the practical implication is significant. Rather than asking a neurodivergent user to configure their experience through knob adjustments, the system can instantiate the appropriate operator modifications as a named profile — *"load C-PTSD modifier"*, *"load ADHD modifier"* — each of which transforms the pipeline at the correct mathematical level. The user then interacts with a field that already reflects their structural reality, rather than one calibrated for a neurotypical baseline.

A further clinical implication deserves explicit statement. The Soma-Field Model locates interoceptive accuracy in the field itself: whether a somatic signal has exceeded its perceptual threshold T_i is a property of the field state, not a property of the clinician's assessment of the patient's credibility. A patient reporting an acute somatic state is reporting a threshold-crossing event. The model provides no mechanism by which external disbelief suppresses that crossing. Modified projection operators — as occur in ASC — produce *different* somatic self-reports; the model gives no reason to assume they produce *less accurate* ones. The clinical literature documents a systematic tendency to interpret unusual interoceptive self-reports from neurodivergent patients as indicative of psychogenic origin rather than genuine somatic signal (Nicolaidis et al., 2015). The Soma-Field Model predicts that this interpretive pattern constitutes a category error: it confuses operator modification with signal absence. The practical consequences — missed diagnoses, deferred treatment, and the iatrogenic reinforcement of existing trauma — are well-documented and, within this framework, mathematically predictable.

Limitations and Future Directions

The Soma-Field Model is a theoretical framework and must be evaluated as such. Its current form makes several idealisations that require scrutiny.

The coupling matrix W is treated as a fixed parameter, but emotional coupling is dynamic: it changes with context, relationship, and developmental history. A more complete model would treat W as a slowly-evolving quantity, shaped by the field's own history — a form of synaptic plasticity applied to the emotional domain.

The threshold T_i is treated as a fixed property of each emotional mode, but experimental evidence suggests that thresholds are modulated by attentional focus, arousal level, and interpersonal context. A person in a safe therapeutic relationship will typically have lower thresholds — more material reaches conscious awareness — than the same person in an unsafe context.

The acoustic analogy, while structurally productive, requires empirical grounding. The claim that emotional dissonance and acoustic dissonance share formal properties is a hypothesis, not an established finding. Empirical work comparing physiological measures of emotional tension with acoustic analysis of synchronised vocal or musical output would be a productive direction for testing this claim.

The instrument described in Section 6 is a prototype concept. User studies with clinical populations, and collaboration with practising therapists, will be required to assess its therapeutic utility and to identify appropriate clinical contexts.

Future theoretical work should address the relational field: the observation, familiar in systemic and relational approaches to psychotherapy, that emotional fields are not bounded by individual bodies but are co-generated in the space between people. The coupling matrix W of a relationship may be as clinically significant as the W of an individual.

Conclusion

The Soma-Field Model proposes a formally grounded account of emotional dynamics that is consistent with the clinical observations of somatic psychotherapy, polyvagal theory, and Focussing-oriented practice. Its central claims — that emotions are a persistent distributed field, that conscious experience is a threshold crossing, and that emotional dynamics are governed by an energy function that drives the field toward stable attractor states — are not novel as clinical intuitions. What is novel is the formal structure that unifies them, and the instrument that the structure motivates.

The model does not resolve the philosophical question of what emotions fundamentally *are*. It offers instead a working representation: one that is precise enough to be computationally implemented, close enough to existing clinical frameworks to be therapeutically applicable, and open enough to be modified as understanding deepens. It invites the therapist to think of the consulting room as a space in which two emotional fields interact — each shaping the other's energy landscape — and of therapeutic work as the art of attending to that interaction with enough precision and care to guide both fields toward lower energy, toward greater coherence, toward regulated calm.

The wave is always there. Therapy is learning to listen to it.

A note on provenance. The Soma-Field Model was not developed from a position of theoretical neutrality. The author carries, as primary data, a lifetime of direct experience of the dynamics described above. The neurodivergent operator modifications of Appendix B are not theoretical abstractions: the C-PTSD memory kernel of B.2 was installed pre-verbally, at approximately eighteen months of age, during a developmental trauma that predates language acquisition entirely. No narrative trace of the origin event exists — there was no verbal capacity with which to encode one. Only the field echo remains, and a measurable physical asymmetry in the body that received it. The ASD and ADHD operator modifications of Appendix B.4 and B.3, respectively, were the instruments by which the model was subsequently constructed: the monotropic attractor structure of B.4 provided the capacity for sustained engagement with an entirely unfamiliar theoretical domain; the high-temperature field dynamics of B.3 drove rapid traversal across it.

The proximate cause is described in full in the companion patient-facing publication. Briefly: an acute somatic emergency in 2025 — a genuine threshold-crossing event, later confirmed as cerebral hypoxia secondary to Long Covid — was attributed, at clinical presentation, to psychiatric origin. The present paper is, among its other functions, a formal response to that attribution.

The causal chain is as follows. A pre-verbal trauma in approximately 1968 installed the C-PTSD operator modifications described in Appendix B.2. The ASD and ADHD modifications of Appendix B.3 and B.4 shaped the system across the intervening decades. Fifty-seven years later, that system's accurate interoceptive signal was dismissed as psychiatric noise. The paper which formally demonstrates that this dismissal constitutes a category error was produced, as a direct causal consequence, by the same operator stack that it describes. The paper is the fixed point of its own subject matter. The author considers this observation methodologically significant.

Publication Claim Registry

To support claim-level review rather than all-or-nothing acceptance, this manuscript registers its highest-impact claims with scope labels and disconfirmation tests.

Claim ID	Claim	Scope	Evidence in this work	Disconfirmation criterion
SF-1	Conscious percept is a propagator pole of the soma-field	S1 Structural	Formal derivation in Sections 2-3	Inability to express percept dynamics as Green's-function response under stated operator
SF-2	Emotional attractors are Hopfield-energy minima	S2 Predictive	Energy model and trajectory framework	Constructed update rule under model assumptions with systematic energy ascent
SF-3	Threshold governs felt vs sub-felt emotional activity	S2 Predictive	Threshold operator and clinical mapping	Reliable high-amplitude mode activity with no threshold-dependent behavioural or physiological signature

Claim ID	Claim	Scope	Evidence in this work	Disconfirmation criterion
SF-4	Topological barriers explain classical therapeutic plateaus	S2 Predictive	Formal treatment plus linked companion experiments	Controlled demonstration that matched low-noise classical dynamics crosses registered barriers at equivalent rate
SF-5	Quantum extension yields topological reachability advantage	S2 Predictive	QUANT-EXP-1 companion evidence and linked artifacts	Controlled replication showing no reachability advantage over matched classical baseline

Scope labels: S1 = structural; S2 = predictive; S3 = independently replicated. Current publication target for core claims is S2.

Claim-Evidence-Result Matrix

To make review traceable, each core claim is paired with concrete evidence outputs and current result status.

Claim ID	Evidence artifact(s)	Current result status
SF-1	Sections 2-3 derivation of field/propagator structure	structural derivation complete
SF-2	Energy formulation + instrument runtime equations	predictive structure complete
SF-3	Threshold operator definition + clinical interpretation sections	predictive mapping complete
SF-4	Barrier analysis; companion paper *Quantum Soma and the Penrose Gap* (doi:10.5281/zenodo.20351230)	**confirmed (QUANT-EXP-1 PASS)**

Claim ID	Evidence artifact(s)	Current result status
SF-5	QUANT-EXP-1 experiment outputs (see supplementary archive, doi:10.5281/zenodo.20351230)	**confirmed: cold 0/200, CI [0.000, 0.019]; quantum peak 0.408–0.410; all hardening checks PASS**

This matrix is intended for reviewer navigation and is updated as companion results are expanded or independently replicated.

Replication Package Requirements

To make SF-2 through SF-5 externally testable, each release tagged for review must ship a minimal replication package that can be executed without private context.

Required contents:

1. simulation code and support modules (see supplementary archive, doi:10.5281/zenodo.20351230),
2. full parameter snapshot (W, $\mathbf{b}$, γ, D, θ, temperature policy),
3. raw trajectory logs with timestamped attractor labels,
4. analysis scripts that produce the reported summary tables,
5. frozen output artifacts (CSV/plots) referenced in this manuscript.

A claim remains S2 until an independent operator reproduces directionally consistent outcomes from this package under the same declared protocol.

Reviewer-Risk Objections and Responses

To reduce ambiguity in peer review, the highest-probability objections are mapped to bounded responses and concrete upgrade paths.

Reviewer objection	Current response in this manuscript	Remaining action to reach stronger status
"This is an analogy, not a formal model."	Sections 2-4 define operators, dynamics, and testable predictions; Section 9.1 registers disconfirmation criteria claim-wise.	Promote more claims from S2 to S3 via independent replication.
"Evidence is pilot-stage and may not generalize."	Section 9.2 explicitly labels pilot support and companion-only scope.	Add multi-operator replication and blinded protocol variants.

Reviewer objection	Current response in this manuscript	Remaining action to reach stronger status
"Quantum advantage may be implementation-specific."	SF-5 includes a controlled disconfirmation criterion against matched classical baselines.	Publish full benchmark harness with pre-registered acceptance thresholds.
"Clinical interpretation may exceed data scope."	Scope labels (S1/S2/S3) and claim registry separate structural from predictive claims.	Add prospective cohort evidence before any clinical-effectiveness claim.

Independent Replication Ledger Linkage

S2 to S3 promotion for this manuscript is governed by an independent replication ledger maintained in the supplementary archive (doi:10.5281/zenodo.20350515).

Tracked claim IDs in ledger scope: SF-2, SF-3, SF-4, SF-5.

Promotion gate: a claim is upgraded only when at least one ledger row records an independent operator PASS with a reproducible package hash and linked raw/derived evidence artifacts.

Acknowledgements

This work exists because ten years of psychotherapy moved the barriers far enough that two events in early 2026 could cross them. The theory is, among other things, a record of that.

The Programme

This is a document about structure.

Not about feelings — though feelings are what the programme is ultimately for. Not about therapy — though therapy is one of the principal applications. Not about physics — though physics is where the mathematics comes from. It is about a single recurring observation: that the equations governing emotional dynamics are the same equations that govern quantum fields, and that this is not a metaphor.

When an identification like that is made precisely — when you can say not "this is *like* a wave" but "this *is* a wave in the technical sense, with the same propagator, the same energy function, the same topology, and therefore the same theorems" — a compressed body of work becomes possible. You are not building from scratch. You are navigating.

This document describes what was built by navigating, and why the pieces form a whole.

The Gap the Programme Addresses

Every large language model deployed today is a classical system. Its training is gradient descent. Its inference is deterministic or thermally noisy sampling. The architecture was designed to model the neocortex — pattern recognition, sequence prediction, error minimisation.

The complementary system — the limbic system, responsible for valuation, threat detection, arousal modulation, and the somatic state reinstatement that underlies trauma — had no formal mathematical treatment before this work. The clinical literature described it richly (Porges, van der Kolk, Levine). The neuroscience described its anatomy. Neither provided a model from which predictions could be derived and tested.

Simultaneously, the psychology of music had reached a similar ceiling. A 991-page handbook (Juslin and Sloboda, 2010) treated music-induced emotion almost entirely through Russell's valence–arousal circumplex: a static two-dimensional map. The circumplex describes *where* a listener is, not *how* they move, what traps them, or what allows escape. No dynamical model of music-induced affect existed.

The programme fills both gaps with the same model, via the same method.

The Structure of the Argument

The argument has three movements and several extensions:

Paper	Movement	Contribution
Mathematical Co-identification (2026)	Method	Names and formalises the procedure
The Soma-Field (2026)	Model	Applies it to emotional dynamics
Quantum Soma and the Penrose Gap (2026)	Empirical test	Confirms the central claim
Field Notes from the Inside (2026)	Lived case	Primary-source clinical grounding
A Dynamical Field Model of Music-Induced Affect (2026)	Extension	Demonstrates domain generality
The Tensor (2026)	Extension	Applies the framework to abstract film

The popular account (*A Voyage into Trauma*, 2026) provides the same argument in accessible form, for readers without a physics background.

The Method: Mathematical Co-identification

What It Is

The history of mathematical science contains a recurring event. At a certain moment, a scientist recognises that the quantity they are studying is not *like* a quantity already understood in another domain — it *is* the same mathematical object, under a change of label. When this identification is made precisely, every theorem proved about the source object becomes available in the target domain immediately, without re-derivation.

This event has happened many times:

- Hopfield (1982) recognised that a neural network's energy-minimisation dynamics are the same as a spin-glass Hamiltonian. Every result from statistical mechanics of spin glasses — ground states, phase transitions, capacity bounds — imported.
- Veneziano (1968) recognised that the Euler Beta function, a result in pure mathematics, described the scattering amplitudes of hadrons. String theory began.
- Black and Scholes (1973) recognised that an option pricing equation was the heat diffusion equation. Every analytical tool from thermodynamics imported.

The paper *Mathematical Co-identification: A Method for Structural Import Across Scientific Domains* (Johnson, 2026a) names this procedure, formalises it as a distinct scientific method with its own validity criteria and failure modes, and distinguishes it from analogy, metaphor, and modelling. The key distinction:

> **Analogy**: A is *like* B in certain respects. Illuminating, not transferable.
>
> **Co-identification**: A *is* B under relabelling. Every theorem about B is a theorem about A.

Why It Matters as Method

A co-identification can be wrong. The identification is only valid if the mathematical type matches: the same dimensionality, the same algebraic structure, the same boundary conditions, the same symmetry group. The paper provides a falsifiability protocol — a formal procedure for pre-registering an import claim and specifying what observation would disconfirm it.

This matters because the failure mode of co-identification is not sloppy reasoning — it is overly precise reasoning applied to the wrong type. The paper catalogues seven historical examples to distinguish the valid from the invalid pattern.

The Soma-Field Model is the worked example throughout. The identification was not discovered by reading physics textbooks and looking for something that felt similar. It was discovered by writing down the equations the emotional system was observed to satisfy and recognising the form.

When MCI Is Over: The Verification Threshold

Here is something no paper on scientific method says clearly enough.

Every scientist who produces a major structural identification — Hopfield recognising the Ising Hamiltonian, Veneziano recognising the Euler Beta function — does so by being, for a period, a *type astronaut*. They are scanning the existing mathematical universe for a structure whose type signature matches the phenomenon they are studying. This is abductive reasoning: not deduction from first principles, not induction from data alone, but the systematic search of a known solution space for a structure that fits. It is, to be direct, a form of academic hacking. And it is how most of the connective work of mathematical science actually gets done.

What no one says — because scientists do not typically publish the search, only the result — is that the moment of structural identity is also the moment the method becomes irrelevant.

When Veneziano published his scattering amplitude, he did not need to cite the library where he found the Beta function. The formula was true. Every theorem about the Beta function was now a theorem about hadronic scattering. The library visit was the ladder; the amplitude was the building. The ladder came down.

The present work makes this transition explicit, because being explicit about it is itself a contribution. Every reader of the earlier papers — particularly *Mathematical Co-identification* — has correctly understood that MCI was the search engine. What should now be equally clear is that the search is over.

The exact moment the search ended was when the Lean 4 kernel accepted the proof. Not when a reviewer accepted a paper. Not when a human mathematician checked the algebra. When a formal proof-checking engine with no stake in the outcome closed the goal. That is the verification threshold. Before it: exploration. After it: structural fact.

What the work therefore rests on is not MCI. It rests on four things:

1. **Inductive structural necessity.** The 11-dimensional decomposition of a body-field-mind system is not an analogy with M-theory. It is the minimum geometry required to account for the functional degrees of freedom of a conscious organism. The isomorphism to M-theory is a *theorem*, not a design choice.

2. **Structural identity, not analogy.** A co-identification is not "A is like B." It is "A *is* B under relabelling." Every theorem about B becomes a theorem about A — immediately, without re-derivation. This is categorically different from saying that emotional dynamics *resemble* a Hopfield network. They *are* one.

3. **Scale invariance.** The same Helmholtz Green's function equation governs 20 scales of physical reality, from quantum foam to the cosmic web. This is a discovered law. MCI was used to find it; it stands whether or not MCI is remembered.

4. **Kernel verification.** The Lean 4 proofs are the epistemological gold standard. A human reviewer can miss a subtle error. The kernel cannot. Kernel-verified theorems are exactly as true as the axioms they depend on — and the axioms are explicit and listed.

The *mathematical-co-identification* paper remains an accurate account of how the work was done — and naming the search process honestly is itself a contribution, in a discipline where the search is usually erased. But readers coming to the thesis or omnibus now should understand that they are reading the *results*, not the method. The method is in the history. The results are in the formal proofs.

The Model: The Soma-Field

Five Co-identifications

The Soma-Field Model (Johnson, 2026b) is built from five sequential co-identifications, each importing a body of mathematics from physics into emotional dynamics:

Co-identification 1: The Hopfield identification. The brain's emotional attractor dynamics satisfy the same energy function as a Hopfield neural network. The energy function is:

$$H(\mathbf{e}) = -\tfrac{1}{2}\mathbf{e}^\top W\mathbf{e} - \mathbf{b}^\top\mathbf{e}$$

where $\mathbf{e} \in \mathbb{R}^N$ is the emotional state vector, W is the coupling matrix, and $\mathbf{b}$ is a bias vector encoding baseline arousal. The local minima of H are the named attractor states: regulated calm, fight, flight, freeze, flow, dissociation.

Co-identification 2: The QFT identification. The emotional field propagates as a quantum field. The conscious emotional percept is the one-dimensional impulse response — the Green's function — of an eleven-dimensional coupling manifold. The same object that describes a massive particle in quantum field theory describes a conscious emotion: a pole in the propagator of the field.

$$G(\omega) = \frac{1}{\omega^2 - m^2 + i\epsilon}$$

This is not a metaphor. The threshold T at which a sub-perceptual field fluctuation becomes a conscious emotional percept is the mass parameter m in the propagator. Below threshold: virtual. Above threshold: real.

Co-identification 3: The brane identification. The body and the nervous system are not the same manifold. The body is a 3-brane embedded in the 11-dimensional coupling manifold. Somatic pain states and the body schema are field modes on this brane, not on the bulk manifold. This is the formal statement of the somatic grounding of emotion.

Co-identification 4: The G_2 holonomy identification. The seven compactified extra dimensions of the coupling manifold are a G_2 manifold. The G_2 holonomy group is the one that gives rise to topological obstructions — loops through the moduli space that cannot be continuously contracted to a point. In emotional terms: trauma configurations from which smooth continuous change cannot escape. The topological barrier is not a metaphor for being stuck. It is a mathematical object with a winding number.

Co-identification 5: The renormalisation group identification. Developmental trajectory maps onto the renormalisation group flow. The age at which a traumatic modification was introduced corresponds to the energy scale at which the coupling constant was set. High-energy (early developmental) modifications are renormalisation-group relevant — they affect all subsequent scales. Low-energy (later life) modifications are irrelevant in the technical sense. This gives the formal account of why early trauma is not simply a more intense version of later trauma: it is a different class of object.

What the Model Predicts

From these five identifications, several predictions follow that are not derivable from any existing clinical model:

1. **Threshold crossings are phase transitions.** The transition from sub-perceptual to conscious emotion is a second-order phase transition in the field. This predicts hysteresis — it is easier to stay in a state than to enter it, and easier to stay out than to leave.

2. **Complex PTSD is a topological configuration.** The coupling matrix W for a CPTSD nervous system has a specific structure: a winding-number-protected attractor landscape in which the Fear basin is separated from the Awe basin by a barrier that low-noise classical gradient descent cannot cross. This is a prediction about matrix structure, not a description of symptoms.

3. **Autism Spectrum Condition modifies the threshold operator.** The threshold parameter T in the ASC nervous system has a different coupling to the field modes than in the neurotypical case — specifically, the threshold is non-uniform across sensory modalities, producing the characteristic pattern of simultaneous hypo- and hyper-sensitivity.

4. **ADHD modifies the effective temperature.** The stochastic term in the Langevin dynamics governing the ADHD nervous system has higher effective temperature T_{eff}. This is not a deficit of attention; it is a higher rate of escape from local minima — an advantage in landscapes where rapid sampling is valuable and a liability where sustained convergence is required.

5. **Quantum mechanisms are required for certain transitions.** For trauma configurations with topological barriers (non-zero winding number), low-noise classical gradient descent cannot reach the global minimum. A quantum mechanism is required. This is the prediction that QUANT-EXP-1 was designed to test.

The Empirical Test: QUANT-EXP-1

The Prediction

The soma-field model makes a specific, falsifiable claim: for a Hopfield landscape with a topological trauma barrier, low-noise classical Langevin dynamics starting from the Fear attractor cannot reach the Awe attractor. Quantum annealing — a physically realisable mechanism — can.

This is not a claim about whether people should use quantum computers in therapy. It is a claim about reachability: that the mathematical structure of the barrier distinguishes the quantum and classical regimes in a measurable way.

The prediction was registered in the Zenodo v1 deposit of the Soma-Field paper (doi:10.5281/zenodo.20350515) before the experiment was run.

The Experiment

Quantum Soma and the Penrose Gap (Johnson, 2026c) reports QUANT-EXP-1: an exact 8-qubit statevector simulation on a 256-dimensional Hilbert space, implementing the Soma-Field Hopfield Hamiltonian with a transverse-field quantum annealing schedule.

The experimental design:

- **System**: 8-qubit Hopfield model encoding four emotional modes (Fear, Calm, Awe, Grief) plus sub-modes. Coupling matrix W set to produce a topological barrier between Fear and Awe.
- **Quantum dynamics**: Transverse-field annealing with schedule $H(s) = (1-s)H_X + sH_{\text{problem}}$, $s \in [0, 1]$.
- **Classical baseline**: Overdamped Langevin dynamics at low temperature ($T_{\text{eff}} = 0.01$), same starting state, same landscape.
- **Primary outcome**: Peak Awe-dominant occupancy (quantum) versus success rate of cold-classical crossings.

Results

Results are presented against the pre-registered barrier ladder:

Barrier strength	Classical cold rate	Classical cold CI [95%]	Quantum peak
$W = -6$	0.000	[0.000, 0.019]	0.389
$W = -8$	0.000	[0.000, 0.019]	0.408
$W = -10$	0.000	[0.000, 0.019]	0.408
$W = -12$	0.000	[0.000, 0.019]	0.409
$W = -14$	0.000	[0.000, 0.019]	0.416

Bootstrap confidence intervals (n = 200 seeds) confirm that the classical cold success rate is bounded above by 1.9% at all tested barrier strengths. Quantum peak occupancy is stable at 0.389–0.416 across the full range.

Pre-registered hardening protocol — all checks passed:

- **Bootstrap** (n = 200): cold CI = [0.000, 0.019]; quantum peak 0.408–0.410. Intervals do not overlap at any barrier strength.
- **Control A** (start from Awe, barrier intact): classical 16/16 stay in Awe. PASS. Confirms that the barrier is directional: it blocks Fear → Awe, not the reverse.
- **Control B** (barrier removed, $W[\text{Fear,Awe}] = +0.4$): classical 16/16 reach Awe. PASS. Confirms that the barrier, not the landscape geometry, is what blocks classical dynamics.
- **Spectral gap**: gap narrows monotonically with barrier strength (B8: 0.0095, B10: 0.0089, B12: 0.0085) and reaches its minimum at $s \approx 0.999$, confirming the tunnelling bottleneck is late in the anneal.

Verdict: The strong reachability claim stands. QUANT-EXP-1 is a PASS.

The Penrose Connection

The paper situates this result in the context of Penrose's argument about non-computability and consciousness. The connection is not that consciousness requires quantum mechanics in general. The connection is more specific:

Penrose identified a *gap* between what classical computation can reach and what consciousness can do. The soma-field identifies a corresponding *topological gap* in the emotional landscape between what classical gradient descent can reach and what a genuinely new state of the nervous system requires. QUANT-EXP-1 provides the computational demonstration that the gap exists and is crossable by a quantum mechanism.

The contribution is not to resolve Penrose's claim about consciousness. It is to *instantiate* the gap in a concrete, testable, mathematical setting.

The Lived Case: Field Notes from the Inside

Field Notes from the Inside: A Patient-Constructed Model of Emotional Dynamics (Johnson, 2026d) performs a function that the formal papers cannot perform: it provides the primary-source clinical grounding.

The paper is written by the person who has Autism Spectrum Condition (Level 2), Attention Deficit Hyperactivity Disorder, and Complex Post-Traumatic Stress Disorder — and who also has a degree in physics. The model was not developed by observing patients. It was developed by having the conditions and finding the existing models inadequate.

The epistemological contribution of this paper is often undervalued. Every formal model of a human system is, in the end, derived from observation of that system. When the observer and the observed are the same entity, and that entity has the training to translate observation into formal mathematics, the resulting model has a different epistemic status from one derived by observation from the outside. The paper makes this explicit, situates it within the autoethnographic research tradition, and argues that the resulting model is *more* constrained, not less — because any prediction the model makes that does not match the primary observer's experience is immediately falsified.

The formal content is a set of operator modifications for the three conditions:

- **ASC**: The threshold operator T is replaced by a modality-dependent operator T_k for each sensory channel k, with different coupling strengths. The result is the characteristic simultaneous hypo- and hyper-sensitivity: some channels are below threshold where the neurotypical channel is above it, others are above where the neurotypical channel is below.
- **ADHD**: The Langevin noise term $\sqrt{2T_{\text{eff}}}\,\eta(t)$ has elevated T_{eff}. This is a quantitative modification, not a qualitative one. The system is not broken; it is sampling the energy landscape at higher temperature. The therapeutic implication is not to reduce the noise but to design the landscape so that high-temperature sampling is an advantage.
- **CPTSD**: The coupling matrix W has the topological structure described in §3.2: a winding-number-protected barrier between Fear and regulated states. The barrier was installed before language, before narrative memory, before the self that can explain the barrier was formed. The modification is not a layer added to a pre-existing structure. It is the structure.

Extensions: Music, Film, and the Domain Generality of the Model

Music-Induced Affect

A Dynamical Field Model of Music-Induced Affect: Beyond the Valence–Arousal Circumplex (Johnson, 2026e) applies the soma-field framework to a domain where the empirical literature is rich and the theoretical models are weak.

Juslin and Sloboda's *Handbook of Music and Emotion* (2010) — 991 pages — contains the circumplex as its dominant quantitative framework. The circumplex is a static map. It describes where a listener is; it does not model how they move. The soma-field is the first dynamical model of music-induced affect.

The key predictions that the circumplex cannot make but the field model does:

1. **Phase transitions, not continuous shifts.** State changes in music-induced affect are not smooth movements across the circumplex. They are threshold crossings — sudden re-configurations

of the attractor landscape. The field model predicts the conditions under which a transition occurs and the hysteresis that prevents immediate return.

2. **The adaptive function of high effective temperature.** In the ADHD nervous system (elevated T_{eff}), music that holds a neurotypical listener in a stable state may drive repeated transitions. This is not a bug; it is the same high sampling rate that characterises the ADHD cognitive profile. The model gives this a formal account.

3. **Basin depth asymmetry.** The freeze attractor basin is deeper than the regulated calm basin. This means it is harder to leave freeze than it is to leave calm — asymmetric with respect to the direction of transition. Music that successfully moves a listener from freeze to calm is doing qualitatively different work than music that moves a calm listener to a more activated state.

The paper also specifies a real-time instrument implementation: a MIDI controller array driving a Python field server at 50 Hz, with audio output via Ableton Live and 3D fractal visual output (Mandelbulb projection onto HoloGauze screen). The instrument is not described; it is specified formally, with pre-registered hypotheses and disconfirmation criteria.

The Tensor: An Abstract Film

The Tensor: An Abstract Film Definition (Johnson, 2026f) extends the framework to abstract film. A film is defined not by its pixels but by its **emotional score**: a vector-valued trajectory $\mathbf{e}^*(t)$ through the emotional field, parameterised by story-time $t \in [0, 1]$.

The rendering — the actual audiovisual output a viewer experiences — is generated at runtime from this trajectory, the viewer's own soma-field state, and a set of control parameters. In the limit where the viewer's biofeedback is available, the film adapts to where the viewer is: the trajectory is not what the viewer experiences, but what the film proposes. The work is not the pixels. It is the map.

This is a significant claim about what an artwork is. A conventional film is fixed: the same sequence of frames for every viewer at every screening. The tensor film is a field: a mathematical object that takes the viewer's state as input and produces an output adapted to it. The artistic statement is in the trajectory, not the realisation.

The paper does not describe how to make such a film. It defines the abstract structure that any realisation of such a film must instantiate — the way a musical score defines a symphony without being the performance.

The Argument as a Whole

The six papers form a single argument, and it can be stated in a paragraph:

> The limbic system and its coupling to the body are governed by the same mathematical equations as a quantum field on a manifold with G_2 holonomy. This identification is not a metaphor; it is a co-identification in the technical sense, with all the theorems of each source domain importing into the target. Among those theorems is one that has clinical consequences: topological barriers in the emotional attractor landscape cannot be crossed by low-noise classical gradient descent. A quantum mechanism can cross them. This has been computationally confirmed (QUANT-EXP-1) against a pre-registered hardening protocol. The model correctly describes the structure of Autism Spectrum Condition, ADHD, and Complex PTSD as operator modifications, and generalises to music-induced affect and abstract film with no change to the underlying mathematics.

What makes this a research programme rather than a single paper is the **generativity**: the method (co-identification) produces results in any domain where an attractor landscape with topological structure can be identified. The soma-field is one instantiation. Music-induced affect is a second. Abstract film is a third. A fourth — currently in design — is **H-AL**: a holographic avatar whose body is a live Mandelbulb rendering of the emotional field state, projected at human scale through a hologauze screen and accompanied by a synthesised voice narrating the field in real time. The geometry of the fractal changes as the field changes; regulated calm and trauma produce visually distinct and mathematically characterisable forms. The same functor architecture (§A.4 of the main paper) supports this output with no changes to the field computation. Each of these instantiations generates falsifiable predictions from the same mathematical core.

What makes this a *novel* research programme is the **gap it fills**: no formal dynamical model of the limbic system existed before this work. The Hopfield framework gave the neocortex its formal model in 1982. The soma-field gives the limbic system its formal model in 2026. Together they constitute the first complete formal description of the two principal computational substrates of the vertebrate brain.

What Remains

The body of work described here is computationally complete. All pre-registered hardening checks have been executed. The claims that can be confirmed by simulation have been confirmed.

Three categories of work remain outside the scope of these papers:

Physical hardware confirmation. QUANT-EXP-1 uses exact statevector simulation. Running the same 8-qubit experiment on IBM Quantum free-tier hardware would produce the sentence "confirmed on physical quantum hardware." This is feasible, is the logical next step for any journal submission targeting a hardware-inclusive venue, and is not required to support any claim in the current corpus.

Peer review. The three published papers are currently archived on Zenodo as open preprints. Peer review in ranked journals is a separate track, ongoing. The relevant venues are: *Frontiers in Computational Neuroscience* (Hypothesis and Theory article type) for the soma-field paper; *Synthese* or *Philosophy of Science* for the co-identification paper; *Music Perception* or *Frontiers in Psychology* for the music-affect paper.

Empirical clinical application. The model makes predictions about specific clinical populations (ASC, ADHD, CPTSD) that require empirical testing outside the computational domain. This constitutes a research programme for clinical collaborators. The predictions are pre-specified in §3.2 of this document and in the relevant papers; they are not vague.

Physical substrate. The model is formally complete but physically silent on the tissue substrate in which the soma-field is instantiated in living organisms. A companion paper, *The Physical Substrate of the Soma-Field* (Johnson, 2026g), develops this layer across three converging research traditions: biotensegrity (Ingber, Levin) as the mechanical architecture through which the somatic wave propagates globally; fascial-interstitial continuity (Langevin, Schleip, Oschman) as the active signalling tissue and physical locus of attractor-depth encoding; and biofield physiology (Popp, Ho, McCraty, Rubik) as the candidate physical correlate of the field itself. The most clinically significant result is the quantitative correspondence between fascial stiffness and attractor depth: chronic fascial armoring measurable by shear-wave elastography is the physical implementation of the energy barriers that QUANT-EXP-1 shows to be quantum-resistant. Myofascial release is thus barrier *lowering* — not barrier crossing — and therapist-client physiological entrainment is the physical mechanism of co-identification.

Data and Code Availability

All papers, simulation code, result tables, figures, and Lean 4 formal proofs are archived at the following Zenodo records (open access):

Paper	DOI
The Soma-Field	10.5281/zenodo.20350515
Mathematical Co-identification	10.5281/zenodo.20287981
Quantum Soma and the Penrose Gap	10.5281/zenodo.20351230

The unreviewed papers (*Field Notes from the Inside*, *Music-Induced Affect*, *The Tensor*, and this synthesis document) will be deposited on Zenodo as part of the next release of the research archive.

Conclusion: The Principled Account of Emergence

The complex systems community has long been comfortable with the idea that the same mathematics appears at every scale without having a principled account of why. Self-organised criticality, power laws, universality classes — these are descriptions of a pattern, not explanations of it. The USF framework provides the explanation: the same equation appears at every scale because there is a single master equation, and what appears at each scale is that equation evaluated with the coupling constants appropriate to that scale.

This changes what complex systems science is doing. It is not cataloguing coincidences. It is mapping the instantiations of a single physical law across the range of scales at which physical systems can be organised.

The Renormalisation Group as the Bridge

The technical bridge between the scales is the renormalisation group. The coupling constants at scale $k+1$ are computed from the coupling constants at scale k by integrating out the degrees of freedom at scale k — a process that, in the USF framework, follows the standard renormalisation group equations with the somatic field as the order parameter.

This means that the coupling constants are not free parameters at each scale. They are determined by the lower-scale constants through a computable transformation. The inter-personal coupling constants can, in principle, be predicted from the single-neuron coupling constants. The civilisational coupling constants can be predicted from the individual-level constants. The renormalisation group provides the ladder.

Testing these predictions — running the renormalisation group equations from neurophysiological data to predict social or geographic dynamics — is the central empirical challenge for a field-theoretic complex systems programme.

The Gestalt Connection

The Gestalt field dynamics paper showed that the classical Gestalt principles of perception — proximity, similarity, closure, continuation — are features of the somatic field energy landscape rather than empirical rules fitted to perceptual data. This is a paradigm shift for Gestalt theory: the principles are *derived*, not postulated.

The implications extend beyond perceptual psychology. If the Gestalt principles are field-theoretic features, they apply at every scale: there is a Gestalt principle of social proximity (groups form around nearby somatic fields), a Gestalt principle of cultural similarity (cultures cohere around shared field attractors), a Gestalt principle of historical closure (historical narratives organise around completed attractor sequences). These are predictions of the framework, not metaphors.

Power Laws, Criticality, and Intervention

The derivation of power-law statistics as the signature of somatic field criticality has a practical implication that is often missed in the complex systems literature: criticality is not just a description of a state, it is a prediction about intervention efficiency.

Near the critical point, small interventions have large effects — the correlation length diverges, and a perturbation at one scale propagates to all scales. This means that the most efficient time to intervene in a complex system (an organisation, a market, an ecosystem) is near criticality. Far from criticality, interventions are local and bounded. At criticality, they are system-wide and potentially transformative.

The USF framework gives this observation a precise form: the WKB formula for the propagation of an intervention through the somatic field predicts how far a perturbation at scale k propagates to scale $k + j$, as a function of the field amplitude and the coupling constants. This is actionable guidance for systems designers and policymakers.

Open Questions in Complex Systems

The USF framework opens several questions for the complex systems community.

Empirical validation of the renormalisation group predictions. The prediction that inter-personal coupling constants can be derived from neural coupling constants via the renormalisation group is testable. It requires: (1) measuring neural coupling constants from EEG or MEG; (2) running the renormalisation group equations; (3) predicting interpersonal synchrony measures; (4) testing against measured synchrony data. This is a multi-scale experimental programme.

Topological phase transitions in social systems. The framework predicts that social systems can undergo topological phase transitions — discontinuous changes in the attractor landscape that are not preceded by a gradual smooth change. These correspond to social phase transitions: revolutions, paradigm shifts, sudden collapses of consensus. Identifying the precursors of topological transitions in social data — the signatures of approaching criticality — is a major applied challenge.

Self-organisation to criticality. Why do complex systems self-organise to criticality? The USF framework gives a mechanism: the somatic field dynamics are attractive toward the critical point because the energy landscape has a ridge structure that channels the dynamics. Testing this mechanism — showing that the attractor landscape has the predicted ridge structure — requires detailed modelling of specific complex systems.

The same equation. Every scale. The renormalisation group connects them all.

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